TAB-PO: Preference Optimization with a Token-Level Adaptive Barrier for Token-Critical Structured Generation

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Abstract

Current offline preference optimization methods are effective for language-model alignment but remain poorly matched to ontology-driven structured prediction, where preferred and rejected JSON often differ by only a few schema-defining tokens. In this low-edit-distance regime, sequence-level objectives such as Direct Preference Optimization (DPO) spread gradient mass across non-critical serialization tokens (*gradient dilution*) and can reduce likelihood on rare preferred schema tokens (*token erosion*). To address these limitations, we first develop a confusion-aware preference-construction strategy combining expert-curated ambiguity patterns with validation-set SFT structured-error modes, producing minimally perturbed, schema-valid negatives for ontology-level decision errors. We then introduce Token-Adaptive Barrier Preference Optimization (TAB-PO), a post-SFT objective for token-critical structured generation with a confidence-gated token-level barrier that anchors under-confident schema tokens. On SciERC, with Llama/Qwen models, TAB-PO improves ontology-critical *semantic-label* and *relational-linking* metrics by 11.59% relative to SFT, wins 100% of comparisons against strongest token-level/sequence-level DPO variants, and surpasses strongest frontier baselines on these metrics by 14.71% relative while improving *textual grounding*.

1 Introduction

Structured prediction lies at the core of many high-value language technologies. In these settings, a model must generate outputs that satisfy explicit structural constraints, respect ontology or schema dependencies, and often ground each decision in supporting evidence from the input [1]. This requirement arises across information extraction, where errors can cause missed or incorrectly normalized evidence [2]; legal document analysis, where errors can introduce compliance risk [3]; and biomedical NLP, where incorrect labels or links can propagate scientific noise [4]. Such errors are costly because they distort downstream evidence, violate task constraints, and weaken decision support [5].

Supervised fine-tuning teaches the model the target schema and valid output format, but many residual errors occur at hard decision boundaries between plausible schema-valid alternatives, such as closely related ontology labels, competing evidence spans, or alternative relational links [6, 7]. Preference optimization is therefore a natural post-SFT stage: it can contrast ambiguous correct and incorrect structured outputs and sharpen the model's choices [8, 9].

Direct Preference Optimization (DPO) has become a widely used approach for post-supervised alignment of large language models, replacing explicit reward modeling and online reinforcement learning with an offline Bradley–Terry objective that increases the relative likelihood of preferred responses over rejected responses [9]. DPO and its variants have been effective for open-ended generation tasks such as instruction following, dialogue, and summarization [8, 10–13]. Ontology-driven structured prediction, however, has a different error geometry. Preferred and rejected outputs are often near-identical serialized records that differ in only a few schema-defining tokens while sharing most JSON scaffolding, field names, and formatting tokens.

This low-separation regime exposes a mismatch between sequence-level preference objectives and token-critical structured correctness. Standard DPO can dilute learning signal across non-critical serialization tokens rather than concentrating it on the sparse tokens that determine semantic labels, textual groundings, or relational links; we refer to this mismatch as *gradient dilution*. It can also improve the aggregate preferred-over-rejected margin while reducing the likelihood of under-confident preferred schema tokens, a failure mode we call *preferred-token erosion*. Recent DPO variants address sequence-level pathologies such as calibration drift, reference dependence, or positive-likelihood degradation [14–17], and token-level variants introduce finer-grained preference signals [18, 19]. However, these methods do not explicitly target the low-edit-distance, ontology-constrained preferences that arise in structured extraction.

Addressing this regime requires preference pairs that reflect realistic ontology-level errors and an objective that focuses optimization on the few tokens that decide structured correctness while still improving the preferred-over-rejected preference margin. Our contributions are as follows:

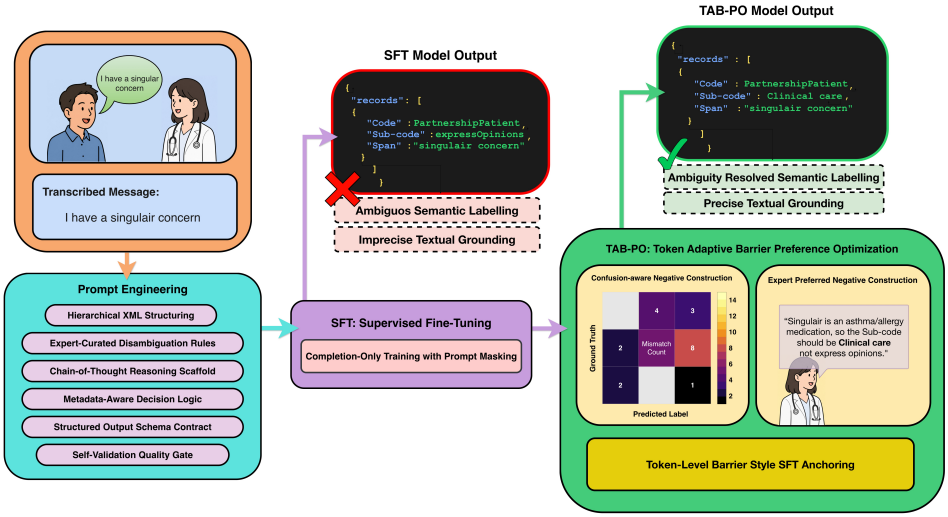


Fig. 1: Overview of the TAB-PO framework for ontology-driven structured prediction. A modular prompt-engineered interface and completion-only supervised fine-tuning teach the model to generate schema-formatted structured records, but residual SFT errors can remain in token-critical decisions such as semantic labelling, relational linking and textual grounding. TAB-PO targets these errors using confusion-aware and expert-guided hard negatives together with a confidence-gated token-level SFT-style likelihood barrier, yielding outputs with resolved semantic and relational ambiguity and more precise source grounding.

- **We introduce a general task formulation for ontology-driven structured prediction.** We define a task-agnostic structured-output setting in which correctness depends on semantic labelling, textual grounding, and relational linking under a task-specific ontology (Section 2).
- **We mathematically characterize why sequence-level DPO is mismatched to low-edit-distance structured preferences.** Through a token-level decomposition of the DPO gradient, we show that standard DPO can allocate update mass to non-critical serialization tokens rather than the sparse tokens that determine structured correctness, leading to *gradient dilution*. We further show that the aggregate preferred-over-rejected margin can improve even when the likelihood of rare or under-confident preferred schema tokens decreases, causing *preferred-token erosion* (Appendix B).
- **We construct confusion-aware structured preference pairs.** We create minimally perturbed, schema-valid hard negatives from empirical SFT ontology-level confusions and expert-adjudicated ambiguity patterns, targeting realistic semantic-label, evidence-grounding, and relational-linking errors (Section 5).
- **We propose Token-Adaptive Barrier Preference Optimization.** TAB-PO augments preference optimization with a confidence-gated SFT-style likelihood

regularizer on under-confident preferred tokens. Together with confusion-aware preferences, TAB-PO reduces gradient dilution, protects vulnerable ontology-bound tokens from likelihood erosion, and preserves the structural scaffolding required for valid serialized outputs (Section 6).

- **We provide comprehensive empirical and diagnostic evidence.** Across PV-Miner and SciERC, and across Llama and Qwen models from 1.5B to 70B parameters, TAB-PO consistently improves over SFT and outperforms strong sequence-level and token-level DPO-family baselines. Mechanistic diagnostics further substantiate these gains by showing that TAB-PO concentrates optimization on schema-critical tokens while maintaining preference separation (Section 7; Section 6.1).

Figure 1 provides an overview of the TAB-PO framework, showing how the modular prompt-engineered interface, supervised structured initialization, confusion-aware preference construction, and token-adaptive barrier objective work together for ontology-driven structured prediction.

2 General Task Formulation for Ontology-Driven Structured Prediction

We introduce a general formulation for ontology-driven structured prediction, covering tasks in which the desired output is not free-form text but a schema-valid set of records constrained by a task-specific ontology. Given an input sequence s, optional task metadata m, and an ontology $\mathcal{O}$, the goal is to predict a finite structured output $\widehat{Y} = \{y_1, y_2, \ldots, y_N\}$. The generation model f_θ maps

$$f_\theta : (s, m, \mathcal{O}) \longmapsto \widehat{Y} = \{y_1, y_2, \ldots, y_N\}. \tag{1}$$

Each predicted record has the abstract form $y_i = (\ell_i, \Pi_i, R_i)$, where the three components correspond to semantic labelling, textual grounding, and relational linking.

Semantic label.

The semantic-label component ℓ_i is drawn from the ontology-defined label space. Depending on the task, ℓ_i may denote an entity type, a relation type, or a hierarchical label tuple. For hierarchical ontologies, we write $\ell_i = (\kappa_i, \varsigma_i)$, where the child label ς_i must be valid under the ontology mapping $\mathcal{H}$, i.e., $\varsigma_i \in \mathcal{H}(\kappa_i)$.

Textual grounding.

Let $\mathcal{S}(s)$ denote the set of admissible evidence spans in the input sequence. The grounding component $\Pi_i \subseteq \mathcal{S}(s)$ contains the span or spans that anchor record y_i to the source text.

Relational linking.

The relational-linking component R_i stores optional typed links from record y_i to other records or grounded mentions. These links encode task-specific inter-record

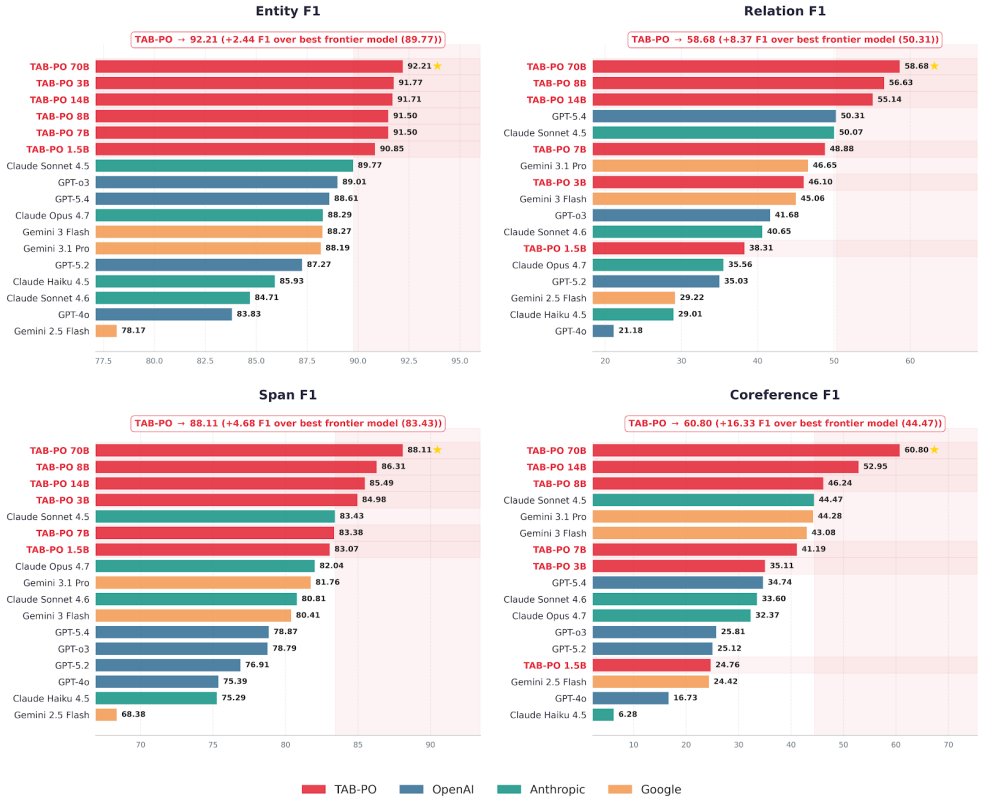


Fig. 2: SciERC F1 comparison between TAB-PO models and frontier models across Entity, Relation, Span, and Coreference extraction. TAB-PO 70B achieves the best performance across all four evaluation dimensions, with particularly large gains on Relation F1 and Coreference F1, the most structurally demanding SciERC tasks. These results demonstrate the effectiveness of token-adaptive barrier preference optimization for ontology-constrained structured prediction.

structure, such as relation arguments, directionality, coreference clusters, or other ontology-defined constraints. If a task does not require relational links, then $R_i = \emptyset$.

The ontology $\mathcal{O}$ determines which semantic labels, grounding spans, and relational links are valid. Concrete task-specific instantiations of this general formulation for PV-Miner and SciERC is presented in Appendix E.

3 Modular Prompt Engineering for Structured Prediction

We next introduce a modular prompt-engineering framework for ontology-constrained structured prediction. The framework is instantiated using replaceable task-specific placeholders, allowing the same prompt architecture to be adapted to any structured

prediction task. It targets common failure modes of in-context structured generation, including format drift, label ambiguity, reasoning shortcuts, evidence hallucination, metadata-conditioned confusion, and invalid relational links.

The prompt decomposes structured prediction into six explicitly controlled modules. **M1** (XML structuring) partitions global instructions and task-specific constraints into semantically tagged blocks that establish scope boundaries for role, performance target, task definition, ontology, and output requirements [20]. **M2** (disambiguation rules) encodes expert-derived decision boundaries for confusable ontology labels, grounding boundaries, and relational-link validity [6, 21]. **M3** (reasoning scaffold) defines a structured verification routine that orders context analysis, candidate decomposition, semantic labelling, textual grounding, relational-link validation, and final consistency checking before output emission [22]. **M4** (metadata-aware decision logic) converts task metadata into explicit control variables that restrict the valid search space for labels, evidence spans, and links [8, 23]. **M5** (schema contract) declares machine-parseable validity constraints for records, labels, groundings, relational links, and task-specific coverage requirements [6, 24]. **M6** (quality gate) defines a single-turn pre-emission audit for parseability, ontology validity, grounding fidelity, relational-link consistency, disambiguation-rule compliance, and expert defensibility [25, 26].

Appendix D provides the detailed component-level explanation of the modular prompt-engineering framework, and Figure D1 in the appendix presents the corresponding general-purpose template.

4 Supervised Structured Initialization

After defining the prompt-engineered structured prediction interface, we perform an initial supervised adaptation stage in which the model learns to emit the corresponding gold schema-valid output under this interface. All subsequent preference-optimization stages build on the resulting supervised model p_{SFT}, which is kept fixed as the reference policy.

Given a gold record set $Y = \{y_1, \ldots, y_N\}$, we serialize it into a JSON completion string Y_s and concatenate it with the instantiated prompt x, where x contains the prompt-engineered instruction populated with the input sequence s, and optional metadata m. Let $z_{1:T} = \mathrm{Tok}(x \,\|\, Y_s)$ denote the tokenized causal-LM training sequence, where $\mathrm{Tok}(\cdot)$ is the model tokenizer and $\|$ denotes string concatenation. Following completion-only instruction tuning, we compute loss only over tokens belonging to the serialized completion Y_s:

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\mathbb{E}_{(x,Y_s)\sim\mathcal{D}_{\mathrm{SFT}}}\left[\frac{\sum_{t=1}^{T} r_t \log p_\theta(z_t \mid z_{<t})}{\sum_{t=1}^{T} r_t}\right]. \tag{2}$$

where $r_t \in \{0, 1\}$ indicates whether token z_t belongs to the completion Y_s. The resulting model p_{SFT} serves as the initialization for preference optimization and is kept fixed as the reference policy in TAB-PO.

5 Confusion-Aware Preference Construction

Having obtained the supervised model p_{SFT}, we next construct preference pairs that expose the model to realistic structured prediction errors. Preference optimization is most effective when the rejected completion represents a plausible failure mode rather than an arbitrary corrupted output. In ontology-driven structured generation, residual SFT errors are often *structured confusions*: the model produces a schema-valid output that is mostly correct but substitutes one plausible ontology-bound decision for another, such as a confusable label, an imprecise evidence span, or an incorrect relational link. These errors are especially challenging because the preferred and rejected outputs may share nearly all serialization tokens and differ only at a small number of task-critical decision points. We therefore construct preference triples (x, Y^+, Y^-), where Y^+ is the gold structured output and Y^- is a minimally perturbed, schema-valid alternative designed to isolate a realistic ontology-level error.

Our construction follows a hybrid protocol. First, we include expert-curated preferences that capture ambiguity patterns identified during annotation review and adjudication (40%). Second, we generate confusion-aware synthetic preferences from empirical errors made by the SFT model on a held-out validation split (60%). This design ensures that preference optimization focuses on realistic ontology-level confusions rather than uninformative random negatives.

Empirical confusion extraction on validation data.

Using the supervised model p_{SFT}, we perform inference on a held-out validation set and compare each predicted structured output $\widehat{Y}$ with the gold structured output Y. Because semantic-label confusions are meaningful only when the corresponding records refer to the same textual evidence, we first align gold and predicted records using relaxed span-level matching. For a predicted span $\hat{\pi} \in \widehat{\Pi}_k$ and a gold span $\pi \in \Pi_j$, let $\mathrm{Tok}(\cdot)$ denote the set of tokens in a span. We use a relaxed span-matching criterion that treats two spans as matched when either span token set contains the other, or when their partial token overlap exceeds a threshold. For the latter case, we compute token-level Jaccard similarity as

$$\mathrm{Jaccard}(\hat{\pi}, \pi) = \frac{|\,\mathrm{Tok}(\hat{\pi}) \cap \mathrm{Tok}(\pi)|}{|\,\mathrm{Tok}(\hat{\pi}) \cup \mathrm{Tok}(\pi)|}. \tag{3}$$

Given a relaxed matching threshold γ, we define

$$\mathrm{Match}_{\Pi,\gamma}(\hat{\pi}, \pi) = \mathbb{I}\big[\mathrm{Tok}(\pi) \subseteq \mathrm{Tok}(\hat{\pi}) \ \vee \ \mathrm{Tok}(\hat{\pi}) \subseteq \mathrm{Tok}(\pi) \ \vee \ \mathrm{Jaccard}(\hat{\pi}, \pi) \geq \gamma\big]. \tag{4}$$

Using this span-matching criterion, we form the set of matched gold–predicted record pairs:

$$\mathcal{M}_\gamma(Y, \widehat{Y}) = \{(j,k) : \exists\, \pi \in \Pi_j, \exists\, \hat{\pi} \in \widehat{\Pi}_k \text{ s.t. } \mathrm{Match}_{\Pi,\gamma}(\hat{\pi}, \pi) = 1\}. \tag{5}$$

In our implementation, we set $\gamma = 0.6$. We then collect mismatched semantic label pairs only over these span-aligned records:

$$\mathcal{A}^{\neq}(Y, \widehat{Y}) = \{(\ell_j, \widehat{\ell}_k) : \ (j,k) \in \mathcal{M}_{0.6}(Y, \widehat{Y}), \ell_j \neq \widehat{\ell}_k\}. \tag{6}$$

Aggregating these mismatches over the validation set yields a frequency-weighted empirical confusion distribution over ontology labels.

Preference triples construction.

The empirical confusion distribution provides a task-specific estimate of the model's most likely ontology-level failure modes. We therefore use it to guide negative construction, ensuring that rejected outputs remain schema-valid while targeting realistic low-separation errors. Each preference triple shares the same input x and differs only in the structured output. We set $Y^+ = Y$ and construct a minimally perturbed Y^- that preserves schema validity while introducing a realistic structured error. We use four different perturbation families:

1. **Semantic confusion with preserved grounding.** This perturbation changes one ontology label while preserving the grounding and relational structure of the selected record.

   Step 1: estimate label vulnerability. Let $C(\ell, \tilde{\ell})$ denote the number of times the SFT model predicts $\tilde{\ell}$ when the gold semantic label is ℓ on the validation set. We define the total confusion count for label ℓ as

   $$c(\ell) = \sum_{\ell' \neq \ell} C(\ell, \ell'). \tag{7}$$

   Step 2: select a vulnerable record. For a structured output $Y^+ = \{y_1^+, \ldots, y_N^+\}$ with $y_i^+ = (\ell_i, \Pi_i, R_i)$, we sample a record index from the vulnerability-weighted distribution

   $$q(i \mid Y^+) = \frac{c(\ell_i) + \epsilon}{\sum_{j=1}^{N} (c(\ell_j) + \epsilon)}, \tag{8}$$

   where $\epsilon > 0$ provides smoothing. This assigns higher probability to records whose labels are more frequently confused by p_{SFT}.

   Step 3: sample a confusable replacement label. Given the selected record $y_i^+ = (\ell_i, \Pi_i, R_i)$, we sample an ontology-valid replacement label $\tilde{\ell}_i$ from the empirical confusion distribution

   $$p_{\mathrm{conf}}(\tilde{\ell} \mid \ell_i) = \frac{C(\ell_i, \tilde{\ell}) + \epsilon}{\sum_{\ell'} (C(\ell_i, \ell') + \epsilon)}. \tag{9}$$

   Step 4: construct the rejected output. We preserve the grounding and relational components and define $\tilde{y}_i = (\tilde{\ell}_i, \Pi_i, R_i)$. The rejected output is

   $$Y^- = Y^+ \setminus \{y_i^+\} \cup \{\tilde{y}_i\}. \tag{10}$$

   Thus, the preferred and rejected outputs differ in a single semantic decision while retaining the same evidence spans and relational links.

2. **Missing-record perturbation.** We delete a record from the preferred output to simulate under-extraction:
$$Y^- = Y^+ \setminus \{y_i^+\}, \tag{11}$$
where y_i^+ is selected according to Eq. 8.
3. **Extra-record perturbation.** We insert a spurious but schema-valid record to simulate over-extraction:
$$Y^- = Y^+ \cup \{\tilde{y}\}, \tag{12}$$
where $\tilde{y} = (\tilde{\ell}, \tilde{\Pi}, \tilde{R})$ is constructed using an ontology-valid semantic label $\tilde{\ell}$, a candidate grounding set $\tilde{\Pi}$ sampled from the source sequence s, and optional relational links $\tilde{R}$, subject to the task-specific ontology, relational-linking rules, and schema-validity constraints.
4. **Relational-link perturbation.**
We modify a relational link to simulate relational-linking confusion:
$$Y^- = Y^+ \setminus \{y_i^+\} \cup \{\tilde{y}_i\}, \tag{13}$$
where $y_i^+ = (\ell_i, \Pi_i, R_i)$ is selected according to Eq. 8, and $\tilde{y}_i = (\ell_i, \Pi_i, \tilde{R}_i)$ preserves the semantic label and grounding spans while replacing R_i with a schema-valid but incorrect relational-link set $\tilde{R}_i$.

For intuitive examples of how these perturbation families produce low-separation, schema-valid hard negatives, see the PV-Miner and SciERC illustrations in Appendix H.

6 Token-Adaptive Barrier Preference Optimization

Motivation.

The preference construction described above yields low-separation structured preference pairs, where preferred and rejected completions often differ in only a small number of schema-defining decisions while sharing most of the same serialized output. Standard sequence-level DPO is poorly matched to this regime. First, its gradients can be diluted across JSON scaffolding and non-critical serialization tokens rather than concentrated on the sparse tokens that determine structured correctness. Second, because DPO optimizes only a relative sequence-level margin, it can improve the aggregate preference margin while reducing the likelihood of some correct preferred tokens. We refer to these effects as *gradient dilution* and *preferred-token erosion*. Appendix B provides the mathematical intuition behind these two failure modes and derives the corresponding token-level decomposition of the DPO gradient.

Objective.

Let Y_s^+ and Y_s^- denote the serialized preferred and rejected completion strings obtained from Y^+ and Y^-. We define the serialized preference dataset as $\mathcal{D}_{\text{pref}}$, whose elements are (x, Y_s^+, Y_s^-). For each serialized completion Y_s, let $u = \text{Tok}(Y_s) =$

Algorithm 1 Token-Adaptive Barrier Preference Optimization

Require: SFT data $\mathcal{D}_{\mathrm{SFT}}$, validation data $\mathcal{D}_{\mathrm{val}}$, expert preferences $\mathcal{D}_{\mathrm{expert}}$, ontology $\mathcal{O}$, relaxed span-matching threshold γ, smoothing constant ϵ, preference temperature β, confidence threshold τ, barrier weight λ

Ensure: TAB-PO policy p_θ

Train p_{SFT} on $\mathcal{D}_{\mathrm{SFT}}$ using completion-only supervised loss $\mathcal{L}_{\mathrm{SFT}}$.

Initialize $p_\theta \leftarrow p_{\mathrm{SFT}}$ and keep p_{SFT} fixed as the reference model.

Confusion-aware preference construction

Run p_{SFT} on $\mathcal{D}_{\mathrm{val}}$, align records with $\mathrm{Match}_{\Pi,\gamma}$, and estimate confusion statistics $C(\ell, \tilde{\ell})$, $c(\ell)$, and $p_{\mathrm{conf}}(\tilde{\ell} \mid \ell)$.

Using $C(\ell, \tilde{\ell})$, $c(\ell)$, and $p_{\mathrm{conf}}(\tilde{\ell} \mid \ell)$, construct $\mathcal{D}_{\mathrm{pref}}$ by combining $\mathcal{D}_{\mathrm{expert}}$ with schema-valid hard negatives generated from semantic-confusion, missing-record, extra-record, and relational-link perturbations.

TAB-PO preference optimization

for each minibatch $\mathcal{B} \subset \mathcal{D}_{\mathrm{pref}}$ **do**

For each preferred completion Y_s^+, compute the detached confidence gate

$$g_t^\theta(x, u^+) = \mathbb{I}\left[\log p_\theta(u_t^+ \mid x, u_{<t}^+) < \log \tau\right].$$

Update p_θ by minimizing the minibatch TAB-PO objective

$$\widehat{\mathcal{L}}_{\text{TAB-PO}}(\theta; \mathcal{B}) = \frac{1}{|\mathcal{B}|} \sum_{(x, Y_s^+, Y_s^-) \in \mathcal{B}} \Bigg[-\log \sigma \Bigg(\beta \Big(\mu_\theta(Y_s^+ \mid x) - \mu_\theta(Y_s^- \mid x) - \mu_{\mathrm{SFT}}(Y_s^+ \mid x) + \mu_{\mathrm{SFT}}(Y_s^- \mid x) \Big) \Bigg) + \lambda \frac{\sum_{t=1}^{T^+} g_t^\theta(x, u^+) \left[-\log p_\theta(u_t^+ \mid x, u_{<t}^+) \right]}{\max\left(1, \sum_{t=1}^{T^+} g_t^\theta(x, u^+)\right)} \Bigg].$$

end for

return p_θ

$(u_1, \ldots, u_T)$ denote its completion-token sequence. We write the completion log-likelihood as

$$\mu_\theta(Y_s \mid x) \triangleq \sum_{t=1}^{T} \log p_\theta(u_t \mid x, u_{<t}). \tag{14}$$

Here, μ_{SFT} denotes the same completion log-likelihood computed under the fixed supervised reference model p_{SFT}. For each preference triple (x, Y_s^+, Y_s^-), we define the reference-adjusted advantage as the difference between the policy log-likelihood margin and the corresponding SFT-reference log-likelihood margin:

$$\Delta_\theta \triangleq \left[\mu_\theta(Y_s^+ \mid x) - \mu_\theta(Y_s^- \mid x)\right] - \left[\mu_{\mathrm{SFT}}(Y_s^+ \mid x) - \mu_{\mathrm{SFT}}(Y_s^- \mid x)\right], \tag{15}$$

The corresponding preference loss is

$$\mathcal{L}_{\text{pref}}(\theta) = -\mathbb{E}_{\mathcal{D}_{\text{pref}}}\left[\log \sigma\left(\beta \Delta_\theta\right)\right]. \tag{16}$$

To address gradient dilution and preferred-token erosion, TAB-PO adds a confidence-gated token barrier on the preferred completion. The barrier activates only when the current policy is under-confident on preferred completion tokens, according to a probability threshold $\tau \in (0, 1)$.

For the preferred serialized completion Y_s^+, let $u^+ = \text{Tok}(Y_s^+) = (u_1^+, \ldots, u_{T^+}^+)$. We define the per-token gate as

$$g_t^\theta(x, u^+) \triangleq \mathbb{I}\left[\log p_\theta(u_t^+ \mid x, u_{<t}^+) < \log \tau\right]. \tag{17}$$

The gate identifies preferred-token positions whose likelihood falls below the confidence threshold. TAB-PO then applies an SFT-style likelihood-restoration penalty only to these activated positions, averaging the preferred-token negative log-likelihood over the selected tokens:

$$\mathcal{L}_{\text{barrier}}(\theta) = \mathbb{E}_{(x, Y_s^+, Y_s^-) \sim \mathcal{D}_{\text{pref}}}\left[\frac{\sum_{t=1}^{T^+} g_t^\theta(x, u^+)\left[-\log p_\theta(u_t^+ \mid x, u_{<t}^+)\right]}{\max\left(1, \sum_{t=1}^{T^+} g_t^\theta(x, u^+)\right)}\right]. \tag{18}$$

When no preferred token is selected by the gate, the numerator is zero and the barrier contributes no loss. In implementation, the binary gate g_t^θ is detached from the computation graph; gradients are propagated only through the selected preferred-token negative log-likelihood terms.

The final TAB-PO objective combines reference-adjusted preference learning with the confidence-gated supervised restoration term:

$$\mathcal{L}_{\text{TAB-PO}}(\theta) = \mathcal{L}_{\text{pref}}(\theta) + \lambda \mathcal{L}_{\text{barrier}}(\theta), \tag{19}$$

where $\lambda \geq 0$ controls the strength of the barrier. Algorithm 1 provides a compact summary of the TAB-PO procedure, showing how validation-set confusion statistics and expert preferences are used to construct schema-valid hard negatives, which are then optimized with the confidence-gated token-adaptive barrier objective.

6.1 Mechanistic Diagnostics

TAB-PO is designed to correct mechanism-level failures that arise when preference optimization is applied to low-separation structured outputs: gradient dilution and preferred-token erosion. We therefore audit the training dynamics directly, rather than relying only on downstream F1. The diagnostics separate completion tokens into three categories: *critical schema tokens*, which determine semantic labels, textual groundings, relation labels, relation arguments, sentence identifiers, or coreference links; *JSON scaffolding tokens*, which encode braces, brackets, commas, colons, and quotation marks; and *non-critical serialization tokens*, such as repeated field names

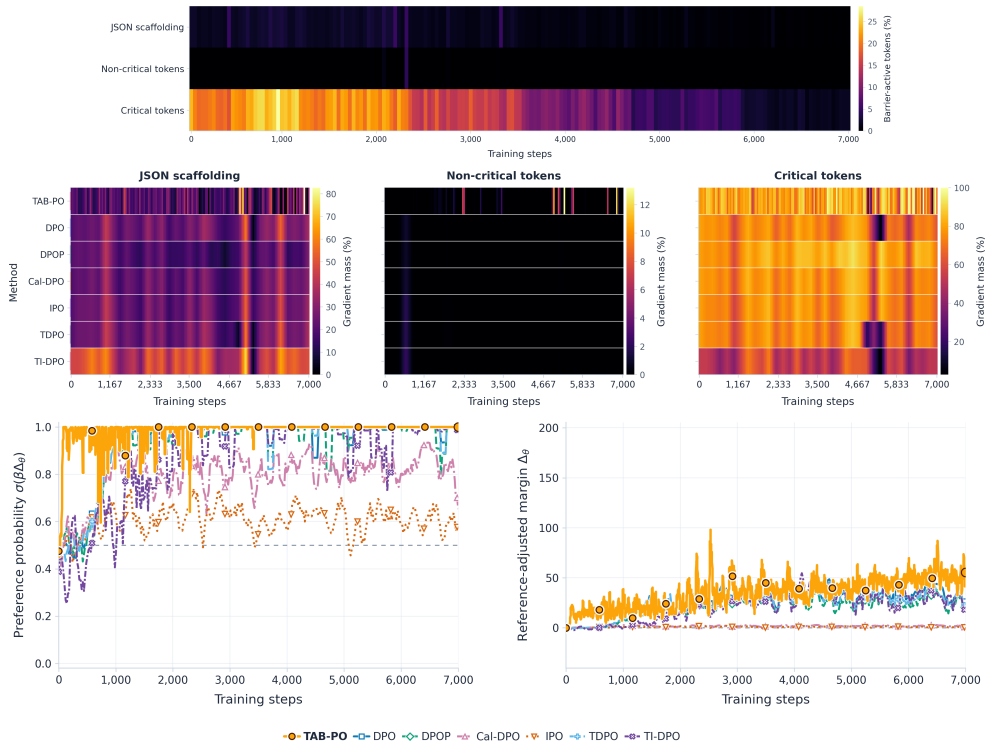


Fig. 3: Mechanistic diagnostics for TAB-PO and preference-optimization variants. The top panel shows TAB-PO barrier activity across JSON scaffolding, non-critical serialization tokens, and critical schema tokens. The middle panel shows gradient-mass allocation across the same token categories for TAB-PO and DPO-family baselines. The bottom panel shows preference dynamics, with preference probability $\sigma(\beta\Delta_\theta)$ and reference-adjusted margin Δ_θ. Together, these diagnostics show that TAB-PO concentrates optimization on schema-critical tokens while maintaining preferred-over-rejected separation.

or other formatting tokens that support parseability but are not themselves scored structured decisions.

Figure 3 reports three complementary diagnostics. Each diagnostic isolates a different part of the proposed mechanism: where the barrier activates, where gradient mass is allocated, and whether preference separation is preserved while vulnerable preferred tokens are restored.

1. **Barrier activity localizes likelihood restoration.** The barrier-activity panel measures the fraction of preferred tokens whose current policy probability falls below the confidence threshold τ and therefore activates the TAB-PO restoration term. TAB-PO activates most strongly on critical schema tokens, especially early in optimization, while remaining weak on non-critical serialization tokens. This shows that the confidence gate is not simply adding uniform SFT regularization;

instead, it selectively restores the token positions where erosion would most affect structured correctness. JSON-scaffolding tokens receive weaker but nonzero activation, which is desirable when structural tokens must be protected to preserve machine-parseable outputs.

2. **Gradient mass shifts toward schema-critical decisions.** The gradient-mass panels examine where preference-optimization updates are allocated. Sequence-level and token-level DPO-family baselines allocate visible update mass to JSON scaffolding and other non-critical serialization tokens. In contrast, TAB-PO shifts a larger share of the learning signal toward critical schema tokens—the sparse positions that determine ontology labels, grounded spans, relation labels, and linking decisions. This directly supports the intended role of TAB-PO in reducing gradient dilution.
3. **Preference separation is preserved without token erosion.** The preference-dynamics panels track both the preference probability $\sigma(\beta\Delta_\theta)$ and the reference-adjusted margin Δ_θ. TAB-PO maintains strong preferred-over-rejected separation while the barrier remains active on under-confident preferred tokens. Thus, the restoration term does not suppress preference learning; instead, it protects vulnerable preferred tokens while the model continues to increase the reference-adjusted preference margin.

Together, these diagnostics show that TAB-PO improves the allocation of optimization signal, preserves token-level stability, and maintains sequence-level preference separation.

7 Experimental Results

Scoring overview.

Figure F1 illustrates the two complementary structured prediction tasks evaluated in this paper, and Appendix F provides the corresponding task-specific scoring definitions. PV-Miner evaluates hierarchical clinical communication mining through coarse Code F1 (parent-level semantic labelling), fine-grained Sub-code F1 (child-level semantic labelling), and grounded Span F1 (textual grounding). SciERC evaluates scientific information extraction through Entity F1 (entity semantic labelling), Relation F1 (typed relational linking), Span F1 (entity textual grounding), and Coreference F1 (cross-mention relational linking). Thus, the reported metrics separate semantic labelling, textual grounding, and relational linking quality.

We first verify that the modular prompt-engineered interface provides a stronger zero-shot initialization before supervised or preference-based fine-tuning. Appendix D reports the full zero-shot comparison between baseline and prompt-engineered instructions in Table D1. Prompt engineering improves nearly all model–metric cells on PV-Miner and all cells on SciERC, supporting its role as an effective structured prediction interface before SFT and TAB-PO.

Table 1: F1 results of TAB-PO compared with recent sequence-level and token-level DPO-family methods on PV Miner and SciERC. Parentheses report the change from the corresponding SFT baseline for the same model and task. Green cells indicate improvements and red cells indicate degradations; darker shading indicates a larger absolute change. All the results are the mean over 3 runs with different random seeds.

(a) PV Miner

Model	Method	Code	Sub-code	Span
Llama-3.1 8B	Cal-DPO	77.43 (↓ 1.67)	74.51 (↑ 0.05)	86.83 (↓ 0.11)
	DPO	79.35 (↑ 0.25)	75.81 (↑ 1.35)	87.24 (↑ 0.30)
	DPOP	79.43 (↑ 0.33)	75.59 (↑ 1.13)	87.28 (↑ 0.34)
	IPO	12.75 (↓ 66.35)	6.25 (↓ 68.21)	6.00 (↓ 80.94)
	TDPO	77.89 (↓ 1.21)	73.77 (↓ 0.69)	85.38 (↓ 1.56)
	TI-DPO	79.80 (↑ 0.70)	75.75 (↑ 1.29)	87.18 (↑ 0.24)
	TAB-PO	82.99 (↑ 3.89)	80.62 (↑ 6.16)	87.67 (↑ 0.73)
Llama-3.2 3B	Cal-DPO	65.51 (↓ 9.61)	65.90 (↓ 3.74)	81.46 (↓ 2.29)
	DPO	74.80 (↓ 0.32)	70.68 (↑ 1.04)	84.48 (↑ 0.73)
	DPOP	74.89 (↓ 0.23)	71.08 (↑ 1.44)	84.25 (↑ 0.50)
	IPO	52.78 (↓ 22.34)	42.69 (↓ 26.95)	49.80 (↓ 33.95)
	TDPO	72.61 (↓ 2.51)	70.14 (↑ 0.50)	82.72 (↓ 1.03)
	TI-DPO	74.88 (↓ 0.24)	70.52 (↑ 0.88)	84.25 (↑ 0.50)
	TAB-PO	80.49 (↑ 5.37)	78.08 (↑ 8.44)	85.29 (↑ 1.54)
Llama-3.3 70B	Cal-DPO	79.77 (↓ 3.92)	75.49 (↓ 4.64)	86.21 (↓ 2.38)
	DPO	83.40 (↓ 0.29)	79.77 (↓ 0.36)	88.60 (↑ 0.01)
	DPOP	83.63 (↓ 0.06)	80.55 (↑ 0.42)	88.40 (↓ 0.19)
	IPO	8.52 (↓ 75.17)	3.21 (↓ 76.92)	11.02 (↓ 77.57)
	TDPO	83.33 (↓ 0.36)	79.70 (↓ 0.43)	88.44 (↓ 0.15)
	TI-DPO	83.67 (↓ 0.02)	80.38 (↑ 0.25)	88.69 (↑ 0.10)
	TAB-PO	84.97 (↑ 1.28)	82.71 (↑ 2.58)	88.03 (↓ 0.56)
Qwen2.5 1.5B	Cal-DPO	65.84 (↓ 7.56)	64.74 (↓ 2.17)	81.97 (↓ 2.32)
	DPO	68.30 (↓ 5.10)	65.36 (↓ 1.55)	84.61 (↑ 0.32)
	DPOP	69.78 (↓ 3.62)	66.66 (↓ 0.25)	84.17 (↓ 0.12)
	IPO	61.43 (↓ 11.97)	51.47 (↓ 15.44)	75.77 (↓ 8.52)
	TDPO	67.74 (↓ 5.66)	64.49 (↓ 2.42)	83.12 (↓ 1.17)
	TI-DPO	72.47 (↓ 0.93)	66.70 (↓ 0.21)	84.63 (↑ 0.34)
	TAB-PO	78.76 (↑ 5.36)	74.01 (↑ 7.10)	85.35 (↑ 1.06)
Qwen2.5 7B	Cal-DPO	74.67 (↓ 2.72)	71.82 (↓ 1.67)	85.64 (↓ 1.31)
	DPO	75.69 (↓ 1.70)	71.62 (↓ 1.87)	85.95 (↓ 1.00)
	DPOP	75.84 (↓ 1.55)	71.99 (↓ 1.50)	86.10 (↓ 0.85)
	IPO	43.44 (↓ 33.95)	32.21 (↓ 41.28)	30.92 (↓ 56.03)
	TDPO	76.01 (↓ 1.38)	71.76 (↓ 1.73)	86.10 (↓ 0.85)
	TI-DPO	76.84 (↓ 0.55)	71.91 (↓ 1.58)	86.18 (↓ 0.77)
	TAB-PO	81.86 (↑ 4.47)	78.31 (↑ 4.82)	87.32 (↑ 0.37)
Qwen2.5 14B	Cal-DPO	77.82 (↑ 2.13)	75.11 (↑ 4.00)	86.91 (↑ 0.95)
	DPO	79.06 (↑ 3.37)	74.88 (↑ 3.77)	87.07 (↑ 1.11)
	DPOP	79.11 (↑ 3.42)	74.91 (↑ 3.80)	87.05 (↑ 1.09)
	IPO	31.86 (↓ 43.83)	33.77 (↓ 37.34)	19.70 (↓ 66.26)
	TDPO	79.12 (↑ 3.43)	74.99 (↑ 3.88)	86.92 (↑ 0.96)
	TI-DPO	79.11 (↑ 3.42)	74.99 (↑ 3.88)	87.11 (↑ 1.15)
	TAB-PO	82.26 (↑ 6.57)	79.50 (↑ 8.39)	87.56 (↑ 1.60)

(b) SciERC

Model	Method	Entity	Relation	Span	Coreference
Llama-3.1 8B	Cal-DPO	90.73 (↑ 0.09)	50.70 (↑ 0.15)	85.38 (↓ 0.02)	41.23 (↑ 0.99)
	DPO	90.93 (↑ 0.29)	51.06 (↑ 0.51)	86.01 (↑ 0.61)	40.64 (↑ 0.40)
	DPOP	91.24 (↑ 0.60)	51.04 (↑ 0.49)	85.91 (↑ 0.51)	41.06 (↑ 0.82)
	IPO	90.50 (↓ 0.14)	50.50 (↓ 0.05)	85.89 (↑ 0.49)	40.91 (↑ 0.67)
	TDPO	90.83 (↑ 0.19)	51.12 (↑ 0.57)	85.94 (↑ 0.54)	39.26 (↓ 0.98)
	TI-DPO	90.32 (↓ 0.32)	49.54 (↓ 1.01)	85.85 (↑ 0.45)	41.56 (↑ 1.32)
	TAB-PO	91.50 (↑ 0.86)	56.63 (↑ 6.08)	86.31 (↑ 0.91)	46.24 (↑ 6.00)
Llama-3.2 3B	Cal-DPO	90.49 (↑ 0.65)	39.19 (↑ 0.54)	82.14 (↓ 1.11)	27.69 (↑ 0.55)
	DPO	90.12 (↑ 0.28)	39.56 (↑ 0.91)	83.36 (↑ 0.11)	30.98 (↑ 3.84)
	DPOP	90.36 (↑ 0.52)	39.81 (↑ 1.16)	83.42 (↑ 0.17)	28.45 (↑ 1.31)
	IPO	90.45 (↑ 0.61)	39.23 (↑ 0.58)	82.94 (↓ 0.31)	30.23 (↑ 3.09)
	TDPO	89.81 (↓ 0.03)	40.88 (↑ 2.23)	83.13 (↓ 0.12)	29.58 (↑ 2.44)
	TI-DPO	89.59 (↓ 0.25)	35.39 (↓ 3.26)	83.21 (↓ 0.04)	26.64 (↓ 0.50)
	TAB-PO	91.04 (↑ 1.20)	46.67 (↑ 8.02)	84.98 (↑ 1.73)	35.11 (↑ 7.97)
Llama-3.3 70B	Cal-DPO	90.83 (↓ 0.12)	54.55 (↑ 1.24)	86.24 (↓ 0.09)	49.72 (↑ 0.14)
	DPO	91.69 (↑ 0.74)	54.51 (↑ 1.20)	86.26 (↓ 0.07)	50.64 (↑ 1.06)
	DPOP	91.22 (↑ 0.27)	52.83 (↓ 0.48)	85.98 (↓ 0.35)	48.40 (↓ 1.18)
	IPO	90.65 (↓ 0.30)	54.34 (↑ 1.03)	86.07 (↓ 0.26)	50.24 (↑ 0.66)
	TDPO	90.99 (↑ 0.04)	53.74 (↑ 0.43)	87.02 (↑ 0.69)	46.72 (↓ 2.86)
	TI-DPO	89.00 (↓ 1.95)	53.82 (↑ 0.51)	86.80 (↑ 0.47)	45.13 (↓ 4.45)
	TAB-PO	92.21 (↑ 1.26)	58.68 (↑ 5.37)	88.11 (↑ 1.78)	60.80 (↑ 11.22)
Qwen2.5 1.5B	Cal-DPO	89.25 (↓ 0.02)	24.11 (↑ 1.60)	67.14 (↓ 0.33)	11.88 (↑ 2.15)
	DPO	89.32 (↑ 0.05)	25.16 (↑ 2.65)	68.85 (↑ 1.38)	12.05 (↑ 2.32)
	DPOP	88.41 (↓ 0.86)	26.95 (↑ 4.44)	70.83 (↑ 3.36)	13.12 (↑ 3.39)
	IPO	89.46 (↑ 0.19)	23.96 (↑ 1.45)	67.59 (↑ 0.12)	11.09 (↑ 1.36)
	TDPO	89.03 (↓ 0.24)	25.38 (↑ 2.87)	68.56 (↑ 1.09)	13.28 (↑ 3.55)
	TI-DPO	89.22 (↓ 0.05)	23.22 (↑ 0.71)	73.54 (↑ 6.07)	10.89 (↑ 1.16)
	TAB-PO	90.85 (↑ 1.58)	38.31 (↑ 15.80)	83.07 (↑ 15.60)	24.76 (↑ 15.03)
Qwen2.5 7B	Cal-DPO	90.21 (↑ 0.63)	41.23 (↑ 0.64)	82.62 (↑ 1.43)	29.99 (↓ 1.28)
	DPO	89.70 (↑ 0.12)	40.40 (↓ 0.19)	81.08 (↓ 0.11)	31.63 (↑ 0.36)
	DPOP	90.23 (↑ 0.65)	42.79 (↑ 2.20)	81.63 (↑ 0.44)	35.66 (↑ 4.39)
	IPO	89.99 (↑ 0.41)	41.21 (↑ 0.62)	80.97 (↓ 0.22)	32.25 (↑ 0.98)
	TDPO	89.60 (↑ 0.02)	40.63 (↑ 0.04)	80.76 (↓ 0.43)	33.68 (↑ 2.41)
	TI-DPO	89.50 (↓ 0.08)	42.97 (↑ 2.38)	83.42 (↑ 2.23)	28.47 (↓ 2.80)
	TAB-PO	91.50 (↑ 1.92)	48.88 (↑ 8.29)	83.38 (↑ 2.19)	41.19 (↑ 9.92)
Qwen2.5 14B	Cal-DPO	91.30 (↑ 0.17)	50.65 (↓ 0.45)	85.31 (↑ 0.70)	45.01 (↑ 2.67)
	DPO	91.07 (↓ 0.06)	49.77 (↓ 1.33)	85.43 (↑ 0.82)	43.41 (↑ 1.07)
	DPOP	90.67 (↓ 0.46)	51.76 (↑ 0.66)	85.48 (↑ 0.87)	46.49 (↑ 4.15)
	IPO	90.85 (↓ 0.28)	50.78 (↓ 0.32)	85.38 (↑ 0.77)	45.49 (↑ 3.15)
	TDPO	91.05 (↓ 0.08)	49.15 (↓ 1.95)	84.60 (↓ 0.01)	46.31 (↑ 3.97)
	TI-DPO	90.93 (↓ 0.20)	50.92 (↓ 0.18)	84.67 (↑ 0.06)	43.60 (↑ 1.26)
	TAB-PO	91.64 (↑ 0.51)	55.14 (↑ 4.04)	85.20 (↑ 0.59)	52.95 (↑ 10.61)

7.1 TAB-PO Consistently Improves over SFT

Table 1 reports the main post-SFT preference optimization results. On PV-Miner, TAB-PO improves over the corresponding SFT baseline by an average of +3.84 F1 across 18 model–metric cells, with the largest gains on the semantic hierarchy: +4.49 Code F1 and +6.25 Sub-code F1 on average.

On SciERC, the improvements are larger. TAB-PO improves over SFT by an average of +5.77 F1 across 24 model–metric cells, with gains concentrated on the most structurally demanding dimensions: +7.93 Relation F1 and +10.13 Coreference F1 on average. Entity F1 and Span F1 also improve by +1.22 and +3.80, respectively.

7.2 TAB-PO Outperforms Sequence-Level and Token-Level DPO Variants

TAB-PO also outperforms recent sequence-level and token-level DPO-family baselines. Against the strongest non-TAB-PO baseline for each model–metric cell, TAB-PO wins

39 of 42 comparisons overall: 17 of 18 on PV-Miner and 22 of 24 on SciERC. Notably, TAB-PO wins all semantic-labelling and relational-linking comparisons, covering PV-Miner Code/Sub-code and SciERC Entity/Relation/Coreference metrics.

The advantage is largest on token-critical structured decisions. On PV-Miner, TAB-PO exceeds the best non-TAB-PO baseline by +4.09 Code F1 and +5.33 Sub-code F1 on average. On SciERC, it exceeds the best non-TAB-PO baseline by +6.01 Relation F1 and +7.07 Coreference F1 on average. These results support the central design of TAB-PO. Broader related-work positioning of TAB-PO with respect to preference-based alignment, DPO-family objectives, token-level preference optimization, and verifier-based RL alternatives is provided in Appendix G.

7.3 TAB-PO Surpasses Frontier Proprietary Models on SciERC

Because PV-Miner is privacy-constrained, frontier model comparisons are conducted only on SciERC. Figure 2 shows that TAB-PO 70B surpasses the strongest frontier proprietary model on every SciERC metric: +2.44 Entity F1, +8.37 Relation F1, +4.68 Span F1, and +16.33 Coreference F1.

The frontier comparison also shows that TAB-PO gains are not limited to the 70B model. Using the macro average across Entity, Relation, Span, and Coreference F1, TAB-PO 14B reaches 71.32 F1, exceeding every frontier baseline and surpassing the best frontier-per-metric macro score of 67.00 by +4.33 F1. Even TAB-PO 3B reaches 64.49 macro F1, surpassing 9 of the 11 frontier baselines, approximately the 82nd percentile of the frontier-model distribution. These results indicate that TAB-PO improves structured prediction quality not only at the largest scale, but also for substantially smaller open-weight models.

Ablation summary.

We further assess whether TAB-PO's gains are attributable to its proposed design components rather than incidental optimization choices. Across PV-Miner and SciERC, performance is strongest when TAB-PO combines confusion-aware hard negatives with expert-preferred chosen outputs. Increasing the number of rejected outputs per input improves ontology-critical metrics, suggesting that exposure to multiple plausible errors provides a stronger preference signal. In contrast, weaker preference construction, less targeted perturbations, overly permissive or overly restrictive confidence thresholds, and reduced barrier strength lead to larger drops. These trends support the central mechanism of TAB-PO: effective preference optimization for structured generation requires both realistic hard negatives and token-level protection for under-confident preferred decisions. The full diagnostic ablation study is reported in Appendix A.

Additional auxiliary loss components for structured prediction, including token weighting, length normalization, and class-balanced reweighting, are described in Appendix C. These components address recurring challenges in structured prediction, including sparse task-critical tokens, variable-length evidence spans, and highly imbalanced ontology label distribution.

8 Discussion

TAB-PO is designed for settings in which preferred and rejected outputs are low-separation structured objects, and where correctness depends on a sparse set of schema-defining tokens. This makes the method especially suitable for ontology-driven structured prediction, a regime we evaluate through two complementary task settings: clinical communication mining in PV-Miner and scientific information extraction in SciERC. These tasks differ substantially in domain, ontology structure, and output dependencies, yet both expose the same underlying optimization challenge: small token-level changes can determine semantic labels, textual grounding, and relational links. The consistency of TAB-PO's gains across these settings suggests that its mechanism is not tied to a single ontology or dataset format, but to a broader class of token-critical structured prediction problems.

A natural direction for future work is to extend TAB-PO beyond structured information extraction to decision-oriented LLM systems in which a small number of tokens can alter downstream actions. Relevant settings include clinical safety triage, legal and regulatory compliance review, insurance claim assessment, quality auditing, decision-support workflows, and tool-calling systems. In such settings, the same low-separation preference geometry can arise: two outputs may be nearly identical in surface form but differ in a decisive label, action, rule interpretation, risk category, or tool argument. Extending TAB-PO to these domains would test whether confusion-aware preference construction and confidence-gated likelihood restoration can improve not only structured prediction accuracy, but also the reliability of LLM-assisted decision-making systems in high-stakes applications.

9 Conclusion

TAB-PO demonstrates that effective preference optimization for structured generation requires objectives and preference pairs that reflect and leverage the geometry of structured correctness. It addresses this setting through two coordinated components: confusion-aware preference construction, which creates schema-valid hard negatives from realistic SFT ontology-level error patterns together with expert-guided ambiguity analysis, and a confidence-gated token barrier, which protects under-confident preferred tokens from likelihood erosion while reducing gradient dilution across non-critical serialization tokens. These mechanisms translate into consistent gains on the dimensions most aligned with token-critical structured decisions. On SciERC, all six model configurations improve both Relation and Coreference F1 over their SFT baselines, with even the smallest gains across models still amounting to +4.04 and +6.00 points, respectively. The benefit holds across model scales from 1.5B to 70B parameters: Qwen2.5-1.5B improves SciERC macro F1 by +12.00 points after TAB-PO, while Llama-3.3-70B improves by +4.91 points. These results show that TAB-PO is a strong fit for ontology-driven structured prediction.

Statements and Declarations

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Ethics approval. This study used de-identified patient–provider secure-message data obtained from participating clinical sites, including Yale New Haven Health System, Cleveland Clinic, and the Texas Association of Charitable Clinics, through approved institutional governance procedures. All procedures were approved by the Yale University IRB and conducted in accordance with relevant laws, institutional policies, and ethical guidelines for the analysis of de-identified clinical data. The project was reviewed under the applicable institutional governance process and determined not to constitute human-subjects research. The privacy rights of individuals were strictly observed, and no directly identifying patient information is reported in the manuscript.

Data and code availability. The SciERC dataset used in this study is publicly available from its original data source. The experiments on SciERC can be reproduced using the public SciERC data together with the code that will be made publicly available. The PV-Miner clinical dataset generated and analyzed during the current study is not publicly available because it is derived from privacy-protected patient–provider secure-message data obtained from Yale New Haven Health System, Cleveland Clinic, and the Texas Association of Charitable Clinics. These data are subject to strict institutional governance, IRB restrictions, data-use agreements, and privacy-protection obligations, and therefore cannot be shared publicly. The code developed for supervised fine-tuning, confusion-aware preference construction, TAB-PO optimization, baseline preference-optimization methods, inference, and metric evaluation will be made publicly available through a public GitHub repository.

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Appendix A Ablation Study

Table A1: Diagnostic ablations for TAB-PO on PV-Miner and SciERC. For PV-Miner, the reference configuration uses one rejected output per input example, Code and Sub-code semantic-label confusions, confusion-aware hard-negative construction with expert-preferred chosen outputs, $\tau = 0.87$, $\lambda = 0.5$, and $\beta = 0.5$. For SciERC, the reference configuration uses one rejected output per input example, Entity and Relation semantic-label/relational-linking confusions, confusion-aware hard-negative construction with expert-preferred chosen outputs, $\tau = 0.87$, $\lambda = 0.5$, and $\beta = 0.5$. Parentheses report absolute F1 change from the corresponding task-specific reference configuration. Green cells indicate improvement and red cells indicate degradation; darker shading indicates larger relative change.

(a) PV-Miner

Diagnostic axis	Setting	Code	Sub-code	Span
Reference	One rejected output pe Code+Sub-code confus confusion-aware+exper $\tau = 0.87$, $\lambda = 0.5$, $\beta =$	80.49	78.08	85.29
A. Preference data preparation diagnostics				
Number of rejected outputs per input example	2	82.42 (↑ 1.93)	79.89 (↑ 1.81)	86.07 (↑ 0.78)
	3	**83.64** (↑ 3.15)	**82.60** (↑ 4.52)	**86.97** (↑ 1.68)
Preference construction	Expert-preferred rejected output	79.24 (↓ 1.25)	76.94 (↓ 1.14)	85.44 (↑ 0.15)
Confusion-aware preference construction	Extra record	80.01 (↓ 0.48)	77.88 (↓ 0.20)	84.59 (↓ 0.70)
	Missing record	80.82 (↑ 0.33)	77.77 (↓ 0.31)	86.50 (↑ 1.21)
B. TAB-PO objective diagnostics				
Barrier confidence threshold	$\tau = 0.25$	61.58 (↓ 18.91)	48.58 (↓ 29.50)	78.81 (↓ 6.48)
	$\tau = 0.50$	78.50 (↓ 1.99)	76.45 (↓ 1.63)	85.01 (↓ 0.28)
	$\tau = 0.95$	80.31 (↓ 0.18)	77.90 (↓ 0.18)	85.72 (↑ 0.43)
Barrier weight	$\lambda = 0.1$	79.13 (↓ 1.36)	76.99 (↓ 1.09)	84.93 (↓ 0.36)
	$\lambda = 2.0$	81.55 (↑ 1.06)	80.81 (↑ 2.73)	86.63 (↑ 1.34)
Preference temperature	$\beta = 0.25$	81.07 (↑ 0.58)	78.11 (↑ 0.03)	85.67 (↑ 0.38)
	$\beta = 0.75$	79.85 (↓ 0.64)	78.33 (↑ 0.25)	86.21 (↑ 0.92)

(b) SciERC

Diagnostic axis	Setting	Entity	Relation	Span	Coreference
Reference	One rejected output pe Entity+Relation confus confusion-aware+exper $\tau = 0.87$, $\lambda = 0.5$, $\beta =$	91.04	46.67	84.98	35.11
A. Preference data preparation diagnostics					
Number of rejected outputs per input example	2	**91.41** (↑ 0.37)	46.90 (↑ 0.23)	84.94 (↓ 0.04)	**37.04** (↑ 1.93)
	3	91.37 (↑ 0.33)	**47.24** (↑ 0.57)	85.33 (↑ 0.35)	36.06 (↑ 0.95)
Preference construction	Expert-preferred rejected output	90.79 (↓ 0.25)	44.29 (↓ 2.38)	84.14 (↓ 0.84)	33.54 (↓ 1.57)
Confusion-aware pair generation	Missing entity	90.81 (↓ 0.23)	46.80 (↑ 0.13)	**85.93** (↑ 0.95)	34.98 (↓ 0.13)
	Extra entity	90.72 (↓ 0.32)	44.89 (↓ 1.78)	84.45 (↓ 0.53)	34.14 (↓ 0.97)
	Extra relation	90.99 (↓ 0.05)	45.65 (↓ 1.02)	84.91 (↓ 0.07)	33.51 (↓ 1.60)
	Extra coreference	90.74 (↓ 0.30)	45.18 (↓ 1.49)	84.23 (↓ 0.75)	33.92 (↓ 1.19)
B. TAB-PO objective diagnostics					
Barrier confidence threshold	$\tau = 0.25$	90.05 (↓ 0.99)	43.13 (↓ 3.54)	83.76 (↓ 1.22)	23.71 (↓ 11.40)
	$\tau = 0.50$	90.31 (↓ 0.73)	43.98 (↓ 2.69)	83.88 (↓ 1.10)	26.87 (↓ 8.24)
	$\tau = 0.95$	90.79 (↓ 0.25)	46.33 (↓ 0.34)	84.62 (↓ 0.36)	34.70 (↓ 0.41)
Barrier weight	$\lambda = 0.1$	90.55 (↓ 0.49)	45.04 (↓ 1.63)	84.64 (↓ 0.34)	33.23 (↓ 1.88)
	$\lambda = 2.0$	91.21 (↑ 0.17)	46.80 (↑ 0.13)	84.50 (↓ 0.48)	35.41 (↑ 0.30)
Preference temperature	$\beta = 0.25$	91.03 (↓ 0.01)	46.31 (↓ 0.36)	85.10 (↑ 0.12)	35.28 (↑ 0.17)
	$\beta = 0.75$	90.34 (↓ 0.70)	45.80 (↓ 0.87)	84.20 (↓ 0.78)	34.73 (↓ 0.38)

This appendix provides an empirical stress test of TAB-PO's proposed mechanism. We separately evaluate its two central design choices: confusion-aware preference construction and confidence-gated likelihood restoration. We then analyze sensitivity to the number of rejected outputs per input, the type of preference construction, the type of confusion-aware perturbation, and key optimization hyperparameters, including the preference temperature, barrier confidence threshold, and barrier strength. Together, these ablations assess whether TAB-PO's gains arise from its intended mechanism rather than from incidental negative-sampling choices or hyperparameter effects.

Table A1 analyzes the components of TAB-PO. Increasing the number of rejected outputs per input example improves PV Miner by +1.51 F1 with two negatives and +3.12 F1 with three negatives on average. On SciERC, the corresponding gains are smaller but consistently positive, with +0.62 F1 for two negatives and +0.55 F1 for three negatives. This suggests that additional hard negatives can improve coverage, but the central gains do not come merely from adding more preference pairs.

The confidence-gated barrier is essential. Lowering the threshold to $\tau = 0.25$ causes large average drops of -18.30 F1 on PV Miner and -4.29 F1 on SciERC, while $\tau = 0.50$

also degrades performance. A weak barrier ($\lambda = 0.1$) reduces performance on both tasks, with average drops of -0.94 F1 on PV Miner and -1.09 F1 on SciERC. These results show that the barrier must activate on the appropriate under-confident preferred tokens with sufficient strength. Conversely, moderate changes to the preference temperature β produce smaller effects, indicating that TAB-PO is less sensitive to the margin temperature than to the confidence-gated anchoring mechanism.

The preference-construction diagnostics further rule out a simple data-only explanation. Replacing the default confusion-aware construction with expert-preferred negative construction reduces average performance by -0.75 F1 on PV Miner and -1.26 F1 on SciERC. Extra-record and missing-record perturbations produce mixed effects, whereas the full confusion-aware construction provides the strongest and most stable reference configuration. These ablations support the intended mechanism: TAB-PO works because it combines realistic ontology-level hard negatives with a token-level barrier that selectively protects under-confident preferred schema tokens.

Appendix B Mathematical Intuition for Gradient Dilution and Preferred-Token Erosion

This appendix gives the mathematical intuition behind two failure modes that can arise when DPO-style preference optimization is applied to low-separation structured outputs: gradient dilution and preferred-token erosion. The main paper reports empirical mechanism-level diagnostics for TAB-PO and DPO-family variants in Section 6.1; here we focus only on why these failure modes arise under DPO.

Given a preference triple, let Y^+ and Y^- denote the preferred and rejected structured record sets for the same input, where each record has the form $y_i = (\ell_i, \Pi_i, R_i)$. Let x denote the input prompt. We serialize the two structured outputs as completion strings Y_s^+ and Y_s^-. Let $u^+ = (u_1^+, \ldots, u_{T^+}^+)$ and $u^- = (u_1^-, \ldots, u_{T^-}^-)$ denote the completion-token subsequences corresponding to Y_s^+ and Y_s^-, respectively. The reference model is the supervised model p_{SFT}.

For a single preference pair, standard DPO minimizes

$$\mathcal{L}_{\mathrm{DPO}}(\theta) = -\log \sigma\left(\beta\left[\log \frac{p_\theta(Y_s^+ \mid x)}{p_{\mathrm{SFT}}(Y_s^+ \mid x)} - \log \frac{p_\theta(Y_s^- \mid x)}{p_{\mathrm{SFT}}(Y_s^- \mid x)}\right]\right), \tag{B1}$$

where $\beta > 0$ controls the preference strength. Define the reference-adjusted preference margin

$$\Delta_\theta(x, Y_s^+, Y_s^-) = \log \frac{p_\theta(Y_s^+ \mid x)}{p_{\mathrm{SFT}}(Y_s^+ \mid x)} - \log \frac{p_\theta(Y_s^- \mid x)}{p_{\mathrm{SFT}}(Y_s^- \mid x)}. \tag{B2}$$

The gradient of the DPO loss is

$$\nabla_\theta \mathcal{L}_{\mathrm{DPO}} = -\beta\sigma(-\beta\Delta_\theta)\Big[\nabla_\theta \log p_\theta(Y_s^+ \mid x) - \nabla_\theta \log p_\theta(Y_s^- \mid x)\Big]. \tag{B3}$$

Using the autoregressive factorization

$$\log p_\theta(Y_s^\pm \mid x) = \sum_{t=1}^{T^\pm} \log p_\theta(u_t^\pm \mid x, u_{<t}^\pm),$$

we obtain

$$\nabla_\theta \mathcal{L}_{\mathrm{DPO}} = -\beta\sigma(-\beta\Delta_\theta)\left[\sum_{t=1}^{T^+} \nabla_\theta \log p_\theta\big(u_t^+ \mid x, u_{<t}^+\big) - \sum_{t=1}^{T^-} \nabla_\theta \log p_\theta\big(u_t^- \mid x, u_{<t}^-\big)\right]. \tag{B4}$$

Let t_0 denote the first completion-token position at which u^+ and u^- differ, or the first position at which one completion terminates. For $t < t_0$, the preferred and rejected completions contain identical tokens under identical autoregressive prefixes, so their gradient terms cancel. The remaining DPO update is therefore determined by tokens at and after the first divergence. The number of canceled pre-divergence tokens is task- and serialization-dependent. When these shared preferred tokens are under-confident under the SFT initialization, standard DPO provides no direct restoration signal because their preferred and rejected gradient contributions cancel exactly. Thus, SFT uncertainty can remain uncorrected even before the first structured divergence. TAB-PO addresses this gap by applying confidence-gated restoration to under-confident preferred tokens, including tokens whose DPO gradient contribution vanishes due to pre-divergence cancellation.

We group these remaining positions into three categories. Critical tokens are tokens that directly determine structured prediction correctness, including ontology labels, grounded evidence spans, relation labels, relation arguments, sentence identifiers, and coreference links. JSON-scaffolding tokens are purely structural serialization tokens, such as braces, brackets, commas, colons, and quotation marks. Non-critical tokens include schema field names and other formatting or repeated tokens that support parseability but are not themselves scored task decisions.

Let $\mathcal{C}^a$ denote the critical token positions and $\mathcal{N}^a$ denote the union of JSON-scaffolding and non-critical token positions for $a \in \{+, -\}$ after the first divergence. The remaining DPO gradient can be decomposed as

$$G_{\mathrm{crit}} = \sum_{t \in \mathcal{C}^+} \nabla_\theta \log p_\theta\big(u_t^+ \mid x, u_{<t}^+\big) - \sum_{t \in \mathcal{C}^-} \nabla_\theta \log p_\theta\big(u_t^- \mid x, u_{<t}^-\big). \tag{B5}$$

$$G_{\text{non}} = \sum_{t \in \mathcal{N}^+} \nabla_\theta \log p_\theta\big(u_t^+ \mid x, u_{<t}^+\big) - \sum_{t \in \mathcal{N}^-} \nabla_\theta \log p_\theta\big(u_t^- \mid x, u_{<t}^-\big). \tag{B6}$$

Thus, the post-divergence DPO gradient can be summarized as

$$\nabla_\theta \mathcal{L}_{\text{DPO}} \propto -\left(G_{\text{crit}} + G_{\text{non}}\right). \tag{B7}$$

In low-separation structured preferences, the number of tokens that directly encode the scored decision is small relative to the number of serialization and non-critical tokens that remain in the completion:

$$|\mathcal{C}^+| + |\mathcal{C}^-| \ll |\mathcal{N}^+| + |\mathcal{N}^-|. \tag{B8}$$

Moreover, after the first divergence, even surface-identical JSON punctuation or schema field names may occur under different autoregressive prefixes, so their gradient terms need not cancel. Standard DPO can therefore allocate update mass to tokens that do not determine annotation correctness. We refer to this gradient-allocation mismatch as *gradient dilution*.

This analysis also exposes a related failure mode: *preferred-token erosion*. Standard DPO optimizes only the relative sequence-level margin between the preferred and rejected completions, and therefore does not impose a lower bound on the likelihood of any individual preferred token. Let $\alpha_t^+(\theta) = \log p_\theta(u_t^+ \mid x, u_{<t}^+)$ and $\alpha_t^-(\theta) = \log p_\theta(u_t^- \mid x, u_{<t}^-)$. Expanding Eq. B2, we have

$$\Delta_\theta = \sum_{t=1}^{T^+} \left[\alpha_t^+(\theta) - \alpha_t^+(\text{SFT})\right] - \sum_{t=1}^{T^-} \left[\alpha_t^-(\theta) - \alpha_t^-(\text{SFT})\right]. \tag{B9}$$

This aggregate margin can increase even when some preferred-token likelihoods decrease. To see this, consider an update from θ to θ', and define $\delta_t^+ = \alpha_t^+(\theta') - \alpha_t^+(\theta)$ and $\delta_t^- = \alpha_t^-(\theta') - \alpha_t^-(\theta)$. Since p_{SFT} is fixed,

$$\Delta_{\theta'} - \Delta_\theta = \sum_{t=1}^{T^+} \delta_t^+ - \sum_{t=1}^{T^-} \delta_t^-. \tag{B10}$$

Thus, for any preferred token k with $\delta_k^+ < 0$, the margin can still increase whenever

$$\sum_{t \neq k} \delta_t^+ - \sum_{t=1}^{T^-} \delta_t^- > |\delta_k^+|. \tag{B11}$$

A successful DPO update therefore need not preserve the absolute likelihood of every correct preferred token. This is especially problematic for ontology-driven structured

prediction: rare or ontology-specific schema tokens are often precisely the tokens on which the SFT model is least confident. The model may improve the relative preference margin while becoming less reliable at generating the correct structured record. Thus, a purely relative sequence-level objective provides no token-level protection for under-confident preferred schema tokens.

The diagnostics in Figure 3 support this mechanism-level interpretation. TAB-PO's barrier activity is concentrated on critical schema tokens, with much weaker activation on JSON scaffolding and non-critical tokens. The gradient-mass heatmaps show that sequence-level DPO variants can allocate substantial update mass to serialization or non-critical tokens, whereas TAB-PO concentrates learning signal on the sparse tokens that determine structured prediction correctness avoiding *gradient dilution*. The preference-probability and reference-adjusted-margin curves further show that TAB-PO maintains strong preference separation while applying targeted restoration to under-confident preferred tokens avoiding *token erosion*.

Appendix C Auxiliary Loss Components for Structured Prediction

TAB-PO targets ontology-driven structured prediction, where most serialized tokens encode schema scaffolding while only a small subset determines semantic labelling, textual grounding, or relational linking correctness. In addition to the core preference objective and confidence-gated barrier, we consider three auxiliary loss components that address recurring pathologies in structured prediction datasets: sparse critical tokens, large variation in completion length, and long-tailed ontology-label distributions.

C.1 Token-Weighted Preference Aggregation

Structured prediction errors are often localized to a few schema-defining tokens: semantic-label tokens, textual-grounding tokens, relation labels and direction pointers. Standard sequence-level aggregation weights these tokens the same as JSON punctuation, field names, and repeated schema scaffolding. Motivated by loss-reweighting methods that increase the influence of informative or under-emphasized training signals [27], we introduce token weights that concentrate the preference signal on tokens that determine structured prediciton correctness.

For each serialized completion Y_s, define token-index sets $\mathcal{T}_{\mathrm{SL}}(Y_s)$, $\mathcal{T}_{\mathrm{TG}}(Y_s)$, and $\mathcal{T}_{\mathrm{RL}}(Y_s)$ for semantic labelling, textual grounding, and relational linking, respectively. The remaining completion tokens are assigned to $\mathcal{T}_0(Y_s)$. We define

$$w_t(Y_s) = w_{\mathrm{SL}}\mathbb{I}[t \in \mathcal{T}_{\mathrm{SL}}(Y_s)] + w_{\mathrm{TG}}\mathbb{I}[t \in \mathcal{T}_{\mathrm{TG}}(Y_s)] + w_{\mathrm{RL}}\mathbb{I}[t \in \mathcal{T}_{\mathrm{RL}}(Y_s)] + w_0\mathbb{I}[t \in \mathcal{T}_0(Y_s)], \tag{C1}$$

where $w_{\mathrm{SL}}, w_{\mathrm{TG}}, w_{\mathrm{RL}} \geq w_0 \geq 0$. The token-weighted completion log-likelihood is

$$\mu_\theta^w(Y_s \mid x) = \sum_{t=1}^{T} w_t(Y_s) \log p_\theta(u_t \mid x, u_{<t}). \tag{C2}$$

Replacing the unweighted sequence likelihood with μ_θ^w gives the weighted reference-adjusted advantage

$$\Delta_\theta^w(x, Y_s^+, Y_s^-) = \Big[\mu_\theta^w(Y_s^+ \mid x) - \mu_\theta^w(Y_s^- \mid x)\Big] - \Big[\mu_{\mathrm{SFT}}^w(Y_s^+ \mid x) - \mu_{\mathrm{SFT}}^w(Y_s^- \mid x)\Big], \tag{C3}$$

where p_{SFT} is fixed. The corresponding preference loss is

$$\mathcal{L}_{\mathrm{pref}}^w(\theta) = -\mathbb{E}_{\mathcal{D}_{\mathrm{pref}}}\left[\log \sigma\big(\beta \Delta_\theta^w(x, Y_s^+, Y_s^-)\big)\right]. \tag{C4}$$

This weighting changes only how token evidence is aggregated; it does not alter preference-pair construction. It is intended to increase the effective learning signal on semantic labels, grounded spans, and relational pointers while reducing gradient dilution from shared serialization scaffolding.

A corpus-level diagnostic over the canonical serialized gold completions further supports this design: critical serialized tokens account for only about 20.0% of PV Miner completion tokens and 13.4% of SciERC completion tokens on average. Thus, most completion tokens are not directly responsible for the scored structured decision, making uniform sequence-level aggregation a poor match for token-critical structured prediction.

C.2 Per-Example Length Normalization

Structured outputs can differ sharply in length. Some examples contain a single short grounded record, whereas others contain many records, long evidence spans, or multiple relation and coreference links. Length effects are a known source of bias in sequence scoring and generation [28, 29]. In structured prediction, this issue is especially prominent: without normalization, long serialized completions can contribute more total gradient mass simply because they contain more completion tokens, thereby biasing the model toward longer outputs and over-emphasizing examples with many records or long evidence spans rather than examples with more difficult structured decisions.

For instance, PV Miner completions range from 35 to 1,140 serialized tokens, a 32.6× difference, while non-empty SciERC completions range from 116 to 3,112 serialized tokens, a 26.8× difference. This variation is driven by the number of extracted records, span length, and relation/coreference structure. We therefore normalize each weighted log-likelihood by its own token-weight mass. To make each preference pair contribute at a comparable scale regardless of serialized length, we normalize each weighted log-likelihood by its own token-weight mass:

$$Z_w(Y_s) = \max\left(\epsilon, \sum_{t=1}^{T} w_t(Y_s)\right), \tag{C5}$$

where $\epsilon > 0$ is a small constant for numerical stability. The length-normalized weighted completion log-likelihood is

$$\bar{\mu}_\theta^w(Y_s \mid x) = \frac{1}{Z_w(Y_s)} \sum_{t=1}^{T} w_t(Y_s) \log p_\theta(u_t \mid x, u_{<t}). \tag{C6}$$

When length normalization is enabled, the reference-adjusted advantage becomes

$$\bar{\Delta}_\theta^w(x, Y_s^+, Y_s^-) = \Big[\bar{\mu}_\theta^w(Y_s^+ \mid x) - \bar{\mu}_\theta^w(Y_s^- \mid x)\Big] - \Big[\bar{\mu}_{\mathrm{SFT}}^w(Y_s^+ \mid x) - \bar{\mu}_{\mathrm{SFT}}^w(Y_s^- \mid x)\Big]. \tag{C7}$$

The length-normalized preference loss is

$$\mathcal{L}_{\mathrm{pref}}^{w,\mathrm{LN}}(\theta) = -\mathbb{E}_{\mathcal{D}_{\mathrm{pref}}}\Big[\log \sigma\Big(\beta \bar{\Delta}_\theta^w(x, Y_s^+, Y_s^-)\Big)\Big]. \tag{C8}$$

The same normalization is applied to the confidence-gated barrier. Let $g_t^\theta(x, u^+) \in \{0, 1\}$ denote the TAB-PO gate for preferred token u_t^+. The weighted, length-normalized barrier is

$$\mathcal{L}_{\mathrm{barrier}}^{w,\mathrm{LN}}(\theta) = \mathbb{E}_{\mathcal{D}_{\mathrm{pref}}}\left[\frac{\sum_{t=1}^{T^+} g_t^\theta(x, u^+) w_t(Y_s^+)\big[-\log p_\theta(u_t^+ \mid x, u_{<t}^+)\big]}{\max\Big(\epsilon, \sum_{t=1}^{T^+} g_t^\theta(x, u^+) w_t(Y_s^+)\Big)}\right]. \tag{C9}$$

When no preferred tokens are gated, the numerator is zero and the barrier contributes zero. This normalization ensures that examples with long spans, many records, or many relational links do not dominate optimization solely through completion length.

C.3 Class-Balanced Example Reweighting

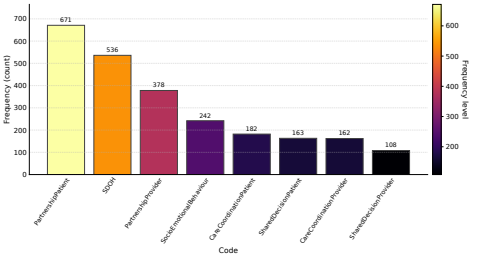
Structured prediction datasets often have long-tailed ontology distributions. This is apparent in the frequency plots for PV Miner Code and Sub-code labels (Figures C1a and C1b) and for SciERC Entity and Relation types (Figure C2). In PV Miner, the most frequent Code, `PartnershipPatient`, appears 671 times, whereas the least frequent Code, `SharedDecisionProvider`, appears 108 times. The Sub-code distribution is more extreme: `Clinical Care` appears 731 times, while `acknowledgePatientExpertiseKnowledge` appears once. In SciERC, `OtherScientificTerm` appears 1,536 times compared with 231 occurrences of `Metric`, and `Used-for` appears 1,687 times compared with 166 occurrences of `Compare`. These skews can cause common labels to dominate minibatch updates and leave rare semantic or relational labels under-optimized.

We therefore apply example-level class-balanced reweighting using the effective-number principle [30]. Let L_{bal} denote the task-specific set of ontology labels used for

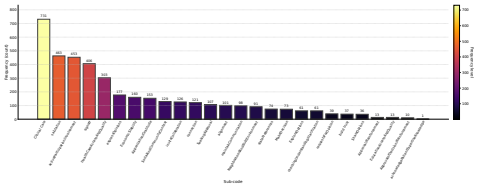
class balancing. For a hierarchical labelling task, this set may include coarse and fine-grained semantic labels. For a relational extraction task, it may include entity labels, relation labels, and, when applicable, coreference or linking decisions.

For each gold structured output $Y^{(i)}$, let $\mathrm{Lab}(Y^{(i)}) \subseteq L_{\mathrm{bal}}$ denote the set of ontology labels selected for class balancing that appear in the labels or typed links of its records. For each such label $\ell \in L_{\mathrm{bal}}$, let n_ℓ be its frequency in the training set. We define

$$\eta(\ell) = \frac{1-\rho}{1-\rho^{n_\ell}}, \qquad \rho \in [0, 1). \tag{C10}$$

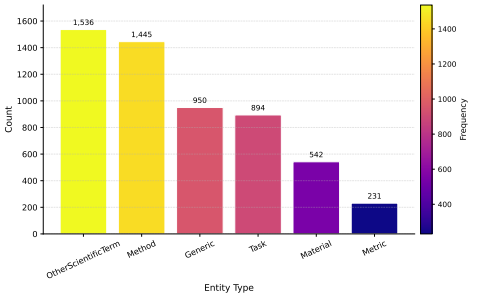


(a) Code-level PV Miner labels. Counts reflect annotation instances rather than unique messages.

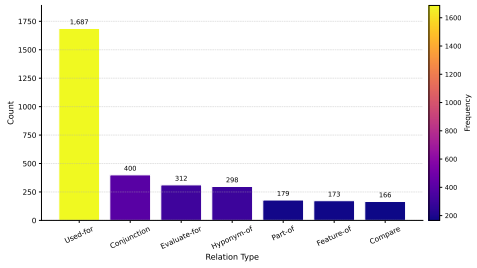


(b) Sub-code-level PV Miner labels. Sub-codes represent fine-grained categories under the Code-level ontology.

Fig. C1: Frequency distributions of PV Miner structured prediction labels across the annotated patient messages.



(a) SciERC entity types. Counts correspond to entity annotation instances in the training split.



(b) SciERC relation types. Counts correspond to relation annotation instances in the training split.

Fig. C2: Frequency distributions of SciERC structured prediction labels in the training split.

Table C1: F1 results of Plain TAB-PO and TAB-PO variants with class balancing, length normalization, and token weighting on PV Miner and SciERC. Cells for the modified TAB-PO settings are colored by the F1 change relative to Plain TAB-PO; green indicates improvement and red indicates degradation, with stronger color intensity indicating a larger absolute change.

(a) PV Miner

Setting	Code	Sub-code	Span
Plain TAB-PO	80.49	78.08	85.29
Plain TAB-PO + Class balancing	81.39	78.77	85.59
Plain TAB-PO + Length normalization	81.79	78.32	86.37
Plain TAB-PO + Token weighting	80.86	79.58	85.69

(b) SciERC

Setting	Entity	Relation	Span	Coreference
Plain TAB-PO	91.04	46.67	84.98	35.11
Plain TAB-PO + Class balancing	91.62	47.21	85.01	35.68
Plain TAB-PO + Length normalization	91.38	46.97	85.63	34.83
Plain TAB-PO + Token weighting	91.44	47.09	85.12	35.81

The example weight is

$$\begin{aligned} \omega_i &= \max_{\ell \in \mathrm{Lab}(Y^{(i)})} \eta(\ell), \\ \omega_i &\leftarrow \min(\omega_i, \omega_{\max}). \end{aligned} \tag{C11}$$

If $\mathrm{Lab}(Y^{(i)}) = \emptyset$, we set $\omega_i = 1$. The max operation emphasizes examples containing at least one rare ontology label, while clipping by $\omega_{\max}$ prevents unstable updates from extremely rare labels.

Let $\mathcal{L}^{(i)}_{\text{TAB-PO}}(\theta)$ denote the per-example TAB-PO loss, including the preference term and confidence-gated barrier, optionally with token weighting and length normalization. For a minibatch of size B, the class-balanced objective is

$$\mathcal{L}_{\mathrm{CB}}(\theta) = \frac{\sum_{i=1}^{B} \omega_i \mathcal{L}^{(i)}_{\text{TAB-PO}}(\theta)}{\sum_{i=1}^{B} \omega_i}. \tag{C12}$$

This normalized weighted average preserves the minibatch loss scale while increasing the influence of examples containing rare semantic labels, rare relation labels, or rare linking decisions.

C.4 Effect of the Auxiliary Components

Table C1 shows that each auxiliary component improves the metric most aligned with its intended role. Class balancing primarily strengthens ontology-level prediction: on PV Miner, Code F1 increases from 80.49 to 81.39 and Sub-code F1 increases from 78.08 to 78.77. Token weighting gives the clearest gain on the fine-grained Sub-code metric, increasing PV Miner Sub-code F1 from 78.08 to 79.58, suggesting that placing more loss mass on schema-critical label tokens helps preserve fine-grained validity mappings. Length normalization most directly improves textual grounding: Span F1 increases from 85.29 to 86.37 on PV Miner and from 84.98 to 85.63 on SciERC. Together, these trends support the intended decomposition: class balancing addresses long-tailed ontology labels, token weighting combats critical-token sparsity, and length normalization reduces optimization bias from variable-length grounded completions.

Appendix D Prompt Engineered Instruction

This appendix provides the design rationale and failure-mode analysis for the six prompt modules (M1–M6) introduced in Section 3. The main text describes the template and its empirical effect; this appendix explains why each module is included and how the modules interact. The template is designed for, and can be used across, a broad class of ontology-constrained structured prediction tasks in which the model must emit schema-valid outputs containing semantic labels, grounded textual evidence, and, when required, typed relational links.

D.1 M1: Hierarchical XML Structuring

M1 organizes the instruction into semantically tagged blocks, including `<role>`, `<performance_target>`, `<task>`, and `<structured_prediction_ontology>`. The model does not parse these tags as formal XML; rather, the tags function as lightweight structural delimiters that separate global behavioral instructions from local task-specific decision rules.

This structure serves two purposes. First, it creates explicit scope boundaries inside a long instruction. Structured prediction prompts often contain role definitions, ontology inventories, disambiguation rules, metadata controls, reasoning steps, and output schemas. Without segmentation, these constraints can interfere with one another or become less salient as the context grows. The tagged structure mitigates this by assigning each type of instruction a clear location and semantic scope. Second, the structure supports modular substitution: a new task can replace the ontology, metadata variables, grounding rules, relational-link inventory, and output schema while preserving the same instruction architecture. This improves portability across structured prediction settings and reduces format drift in long-context generation [20].

D.2 M2: Expert-Curated Disambiguation Rules

M2 encodes expert-derived decision boundaries for cases that are likely to be confused by an instruction-tuned model. These rules may distinguish confusable semantic labels, specify textual-grounding boundaries, define when a candidate Span is too broad or too narrow, or determine whether a relational link is valid. For relational tasks, the rules can additionally specify constraints over the structured relation fields, such as valid evidence pairings, relation type, link directionality, scope, and metadata-dependent validity conditions.

The purpose of this module is to operationalize the annotation manual inside the prompt. Rather than relying on the model to infer boundary conditions from label names alone, M2 states the decision criteria explicitly. This is important because structured prediction evaluation is often sensitive to small local errors: a near-correct label, an over-extended Span, an invalid relation type, an incorrect evidence pairing, or an incorrect sentence references in a coreference record can change the scored output. Crisp disambiguation rules therefore reduce ontology-boundary, grounding-boundary, and relational-linking errors [6, 21].

Table D1: Zero-shot F1 results under baseline (Base) and prompt-engineered instructions (Prompt) on PV Miner and SciERC. F1 cells under Prompt are colored by the relative F1 change from baseline to prompt-engineered instruction. Stronger color intensity indicates a larger relative change.

(a) PV Miner

Model	Code		Sub-code		Span	
	Base	Prompt	Base	Prompt	Base	Prompt
Llama-3.3-70B	60.11	**62.25**	36.13	**43.71**	55.04	**60.86**
Llama-3.1-8B	0.00	**47.09**	0.00	**20.84**	55.24	**54.15**
Qwen2.5-7B	31.71	**42.87**	22.43	**25.78**	40.29	**46.72**
Qwen2.5-14B	56.60	**60.60**	28.80	**37.74**	44.68	**47.55**
Llama-3.2-3B	22.96	**38.24**	10.54	**13.22**	35.29	**38.98**
Qwen2.5-1.5B	19.20	**22.64**	1.95	**17.85**	16.25	**21.16**

(b) SciERC

Model	Entity		Relation		Span		Coreference	
	Base	Prompt	Base	Prompt	Base	Prompt	Base	Prompt
Llama-3.3-70B	78.59	**81.08**	14.82	**16.13**	59.09	**65.10**	0.29	**1.72**
Llama-3.1-8B	73.95	**74.62**	5.64	**10.09**	53.72	**60.71**	0.00	**0.44**
Qwen2.5-7B	70.23	**78.68**	10.90	**13.33**	58.17	**66.60**	0.88	**1.01**
Qwen2.5-14B	78.83	**79.29**	12.86	**18.82**	64.44	**68.01**	5.08	**6.55**
Llama-3.2-3B	27.04	**74.16**	0.88	**3.27**	31.44	**45.84**	0.00	**0.09**
Qwen2.5-1.5B	62.21	**68.17**	4.35	**5.39**	29.98	**41.56**	0.07	**0.31**

D.3 M3: Reasoning Scaffold as Structured Verification

M3 provides a task-specialized reasoning scaffold. The scaffold decomposes prediction into context and metadata analysis, candidate-unit decomposition, semantic label matching, textual grounding with boundary verification, relational linking with structured-field validity verification, and final cross-validation before output emission. Unlike generic chain-of-thought prompting, the scaffold is not intended to elicit free-form explanation; it is a verification routine aligned with the fields of the structured output.

This module addresses reasoning shortcuts and incomplete record construction. A model may identify a plausible label without verifying the evidence Span, or it may extract valid units without checking whether the relation type, evidence, and associated sentence identifier fields are jointly valid under the task schema. M3 forces these decisions to be considered in sequence: first identify candidate units, then assign labels, then verify grounding, then validate relational-link fields, and finally check consistency across the full output. This makes the model less likely to emit records that are syntactically valid but semantically or relationally inconsistent [22].

D.4 M4: Metadata-Aware Decision Logic

M4 exposes task metadata as explicit control variables. Depending on the task, these variables may include speaker role, message direction, sentence identifier, document identifier, section type, temporal context, modality, or other task-specific controls. The key design choice is to make metadata visible to the model as a constraint rather than leaving it as implicit context.

This module narrows the valid search space for structured prediction. Metadata can determine which labels are admissible, which Spans are in scope, which sentence identifier values are valid, and which relational records are possible under the declared schema. By converting latent contextual attributes into observed control signals, M4 reduces metadata-conditioned confusion and prevents plausible but invalid predictions. Its advantage is especially clear in tasks where two records may look lexically similar but differ because of speaker role, sentence context, document section, or relational structure [8, 23].

D.5 M5: Structured Output Schema Contract

M5 specifies the output as a machine-parseable schema contract. The contract defines the required record fields, the representation of grounded evidence, semantic-label validity constraints, relational-link validity constraints, and task-specific coverage requirements. Depending on the task, the schema may specify fields for semantic labels, grounded evidence units, record identifiers, typed links between records or spans, sentence- or document-level provenance, and any additional task-specific attributes required for valid structured prediction.

The schema contract has both modeling and evaluation advantages. At inference time, it reduces format drift by giving the model a fixed structure to fill rather than an unconstrained response format. At evaluation time, it enables deterministic parsing and reproducible metric computation. M5 therefore protects the interface between model generation and automatic evaluation [6, 24].

D.6 M6: Single-Turn Self-Validation Quality Gate

M6 adds a compact validation checklist immediately before final output emission. The checklist verifies that the output is parseable under the declared schema, that semantic labels are drawn from the allowed ontology, that grounded evidence satisfies the task requirements, that relational records satisfy the required constraints, that relevant disambiguation rules have been applied, and that the final output is defensible under expert review.

This module is an intra-generation audit rather than an iterative self-refinement procedure. The model is not asked to perform multi-turn critique-and-revision; instead, it is asked to check concrete validity conditions before emitting the final answer. This distinction is important because single-pass inference is more efficient and more reproducible, and because self-correction without external feedback can be unreliable on reasoning-intensive tasks. M6 therefore functions as a pre-submission audit: it does not add new information, but it increases the likelihood that the generated structure satisfies the constraints already declared in M1–M5 [25, 26].

D.7 Interactions Between Modules

The six modules are designed to be individually interpretable but jointly effective. M1 provides the structural anchors that make the other modules easier to localize and apply. M2 supplies expert boundary conditions that are invoked during the M3 reasoning scaffold. M4 narrows the search space before label, grounding, or relational-link decisions are made. M5 converts those decisions into a strictly parseable schema. M6 then audits the final output against the preceding constraints.

The modules therefore address complementary failure modes. M1 and M5 reduce format drift and schema violations. M2 and M4 reduce ontology ambiguity, grounding-boundary errors, and metadata-conditioned confusion. M3 reduces reasoning shortcuts by separating semantic labelling, textual grounding, and relational-link validation into explicit verification steps. M6 reduces verification omissions by requiring a final single-turn consistency check. The cross-module advantage is that the prompt does not rely on any single mechanism: structural organization, expert rules, metadata

control, stepwise reasoning, schema enforcement, and self-validation work together to constrain the model toward valid structured outputs.

D.8 Prompt Engineering Improves Zero-Shot Structured Prediction

Table D1 shows that the prompt-engineered instruction substantially improves zero-shot structured prediction before any supervised or preference-based fine-tuning. On PV Miner, the mean F1 across all model–metric cells increases from 29.85 to 39.01, corresponding to a +9.17 F1 gain and a +30.72% relative macro improvement. Prompt engineering improves 17 of 18 PV Miner cells, with the largest average gains on ontology-level labels: +13.85 Code F1 and +9.88 Sub-code F1, compared with +3.77 Span F1.

The same trend holds on SciERC. The mean F1 increases from 30.98 to 36.71, corresponding to a +5.73 F1 gain and a +18.50% relative macro improvement. Prompt engineering improves all 24 SciERC model–metric cells. The largest average gains occur for Entity F1 (+10.86) and Span F1 (+8.50), while Relation F1 improves by +2.93 and Coreference F1 improves by +0.63. These results show that structured prompting is an important initialization step: it improves schema adherence and task decomposition, but it does not by itself solve the harder post-SFT preference-optimization problem addressed by TAB-PO.

D.9 Confusion-Matrix Evidence for Prompt-Engineered Instruction

Table D1 quantifies the performance improvements produced by the prompt-engineered instruction over the baseline instruction. Importantly, the baseline is not an uninformed prompt: it contains the essential task definition, semantic-label inventory, grounding requirements, relational-linking requirements where applicable, and input/output schema needed for structured prediction. The gains therefore do not simply come from adding missing task information. Instead, they support the value of organizing the same task knowledge into explicit modules for ontology salience, disambiguation, metadata-aware control, stepwise verification, schema enforcement, and self-validation.

The confusion matrices in Figures D2, D3, and D4 explain where these gains arise. At the Code level, prompt engineering reduces direction-conditioned confusions between patient- and provider-oriented categories. For example, confusions between `PartnershipPatient` and `PartnershipProvider` decrease after prompt engineering. This pattern supports the role of M4, which exposes metadata and control variables as explicit constraints that narrow the valid label search space, and M6, which checks whether the final prediction is consistent with those constraints.

At the Sub-code level, prompt engineering reduces common boundary errors among semantically adjacent communication functions. The baseline shows prominent confusions such as `signoff` → `Appreciation/Gratitude` and `salutation` → `signoff`; after prompt engineering, these confusions decrease. These reductions are consistent with M2, which encodes expert-curated decision boundaries for confusable labels, and M3,

which forces candidate units to be decomposed, labelled, grounded, and cross-checked before output emission.

The SciERC entity-level confusion matrices show a similar effect for scientific information extraction. Prompt engineering reduces several high-frequency off-diagonal confusions, including `OtherScientificTerm` → `Material`, `OtherScientificTerm` → `Method`, `Method` → `OtherScientificTerm`, and `Task` → `Method`. These improvements suggest that the prompt modules help the model separate ontology-near entity types whose surface forms are often lexically ambiguous. They also support the combined benefit of M1, M2, M3, M5, and M6: structural segmentation improves ontology salience, expert rules sharpen local decision boundaries, the reasoning scaffold separates candidate identification from label assignment, the schema contract enforces valid structured output, and the quality gate encourages final constraint checking.

Prompt-Engineered Instruction Template for Structured Prediction

M1 | Hierarchical XML Structuring

<role> [Domain expert persona and behavioral prior] </role>
<performance_target> [Precision-oriented accuracy expectations for semantic labelling, textual grounding, relational linking, and schema-valid structured prediction] </performance_target>
<task> [Task definition, input/output specification, and structured prediction ontology constraints] </task>
<structured_prediction_ontology> [Authoritative ontology with operational definitions for semantic labels, textual grounding units, and relational link types/validity conditions, where applicable] </structured_prediction_ontology>

Separates global instructions from local decision rules; improves constraint salience in long contexts.

M2 | Expert-Curated Disambiguation Rules

<disambiguation_rules>
[Boundary condition 1: decision criteria for confusable semantic label pair A vs. B]
[Boundary condition 2: textual grounding boundary, positional, specificity, or semantic cues]
[Boundary condition 3: relational linking validity, source/target role, directionality, or scope conditions]
. . .
[Boundary condition K: metadata-dependent or context-dependent distinctions]
</disambiguation_rules>

Operationalizes the annotation manual inside the prompt; resolves ontology boundary errors using expert-curated ambiguity adjudication from the annotation process.

M3 | Chain-of-Thought Reasoning Scaffold

<reasoning_process>
Step 1: Context and metadata analysis
Step 2: Candidate unit decomposition and semantic label matching
Step 3: Textual grounding with boundary verification
Step 4: Relational linking with argument-validity verification
Step 5: Cross-validation with loop-back conditions
</reasoning_process>

Imposes a disciplined internal verification sequence before structured output emission.

M4 | Metadata-Aware Decision Logic

<metadata_control>
[Expose task metadata as explicit control variables, e.g., speaker role, message direction, sentence identifier, document identifier, section type, temporal context, source/target role, modality, and task-specific control fields]
</metadata_control>

Converts metadata values, which are latent contextual attributes, into observed control signals; narrows the valid search space for semantic labelling, textual grounding, and relational linking.

M5 | Structured Output Schema Contract

<output_format>
[Machine-parseable hard constraints: schema for structured predictions, grounded evidence, semantic label validity, relational linking validity, and task-specific coverage conditions]
</output_format>

Shifts the task from open-ended generation to structured generation with explicit validity conditions.

M6 | Self-Validation Quality Gate

<quality_gate>
[1. Output is parseable under the declared schema]
[2. Every semantic label is drawn from the allowed ontology]
[3. Every relational link satisfies source/target argument-type constraints]
. . .
[K-1. All disambiguation rules have been applied]
[K. The final output is high-confidence and defensible under expert review]
</quality_gate>

Single-turn intra-generation audit; constrains the model to verify concrete failure conditions within one pass.

Fig. D1: General modular prompt template for ontology-constrained structured prediction with semantic labelling, textual grounding, and relational linking.

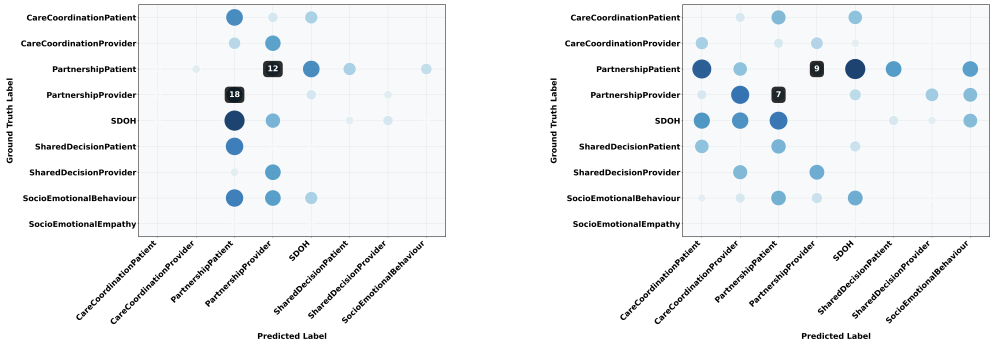


Fig. D2: Code-level confusion matrices comparing baseline instruction outputs (left) and prompt-engineered instruction outputs (right) from Llama-3.2-3B-Instruct.

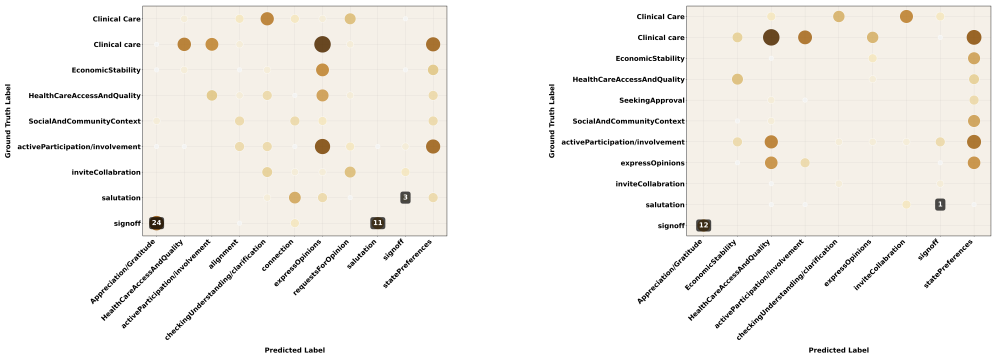


Fig. D3: Sub-code-level confusion matrices comparing baseline instruction outputs (left) and prompt-engineered instruction outputs (right) from Llama-3.2-3B-Instruct. Bubble size indicates the mismatch count between ground-truth and predicted Sub-codes. Only the top 10 most frequently confused Sub-codes are shown for clarity.

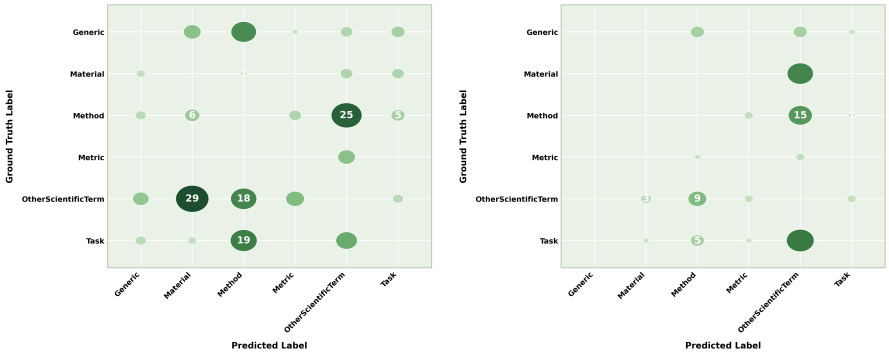


Fig. D4: Entity type-level confusion matrices comparing baseline instruction outputs (left) and prompt-engineered instruction outputs (right) from Llama-3.2-3B-Instruct.

Appendix E Task-Specific Instantiations of the Structured Prediction Formulation

This appendix instantiates the general ontology-driven structured prediction formulation introduced in Section 2 for the two tasks studied in this work.

E.1 PV-Miner

PV-Miner requires the model to extract communication-pattern records from a patient–provider message. The objective is to predict one or more records, where each record contains a coarse-grained Code, a valid fine-grained Sub-code under that Code, and a text Span that grounds the prediction in the source message. Therefore, PV-Miner instantiates the general formulation as hierarchical semantic classification with direct textual grounding and no inter-record relations.

Let $d \in \{\texttt{Y}, \texttt{N}\}$ denote the message-direction metadata, where $d = \texttt{Y}$ denotes provider→patient and $d = \texttt{N}$ denotes patient→provider. Let L_{Code} be the set of valid Codes and $L_{\text{Sub-code}}$ be the set of valid Sub-codes. The PV-Miner ontology defines a hierarchy

$$\mathcal{H}_{\text{PV}} : L_{\text{Code}} \to 2^{L_{\text{Sub-code}}}, \tag{E1}$$

where $\mathcal{H}_{\text{PV}}(\kappa)$ gives the set of Sub-codes valid under Code κ.

The model maps

$$f_\theta : (s, d, \mathcal{O}_{\text{PV}}) \longmapsto \widehat{Y}_{\text{PV}}. \tag{E2}$$

Record Instantiation. Each PV-Miner record is instantiated as

$$y_i^{\text{PV}} = \big((\kappa_i, \varsigma_i), \{\pi_i\}, \emptyset\big), \tag{E3}$$

where

$$\kappa_i \in L_{\text{Code}}, \qquad \varsigma_i \in \mathcal{H}_{\text{PV}}(\kappa_i), \qquad \pi_i \in \mathcal{S}(s). \tag{E4}$$

Thus,

$$\widehat{Y}_{\text{PV}} = \Big\{ \big((\kappa, \varsigma), \{\pi\}, \emptyset\big) \;\Big|\; \kappa \in L_{\text{Code}}, \varsigma \in \mathcal{H}_{\text{PV}}(\kappa), \pi \in \mathcal{S}(s) \Big\}. \tag{E5}$$

Here, $\ell_i = (\kappa_i, \varsigma_i)$ is the ordered hierarchical label tuple, $\Pi_i = \{\pi_i\}$ is the grounding span, and $R_i = \emptyset$ because PV-Miner records do not point to other records.

E.2 SciERC

SciERC requires the model to extract a relational information graph from a scientific document. The objective is to predict typed entity records grounded in text, typed relation records connecting entity records, and coreference cluster records grouping entity records that refer to the same underlying scientific concept. Thus, SciERC instantiates the general formulation as a record graph with grounded entity records and structural relation/coreference records. The model maps

$$f_\theta : (s, \mathcal{O}_{\text{SciERC}}) \longmapsto \widehat{\mathcal{Y}}_{\text{SciERC}}. \tag{E6}$$

The predicted SciERC output is partitioned into three record subsets:

$$\widehat{Y}_{\text{SciERC}} = \mathcal{E} \,\dot\cup\, \mathcal{R} \,\dot\cup\, \mathcal{C}, \tag{E7}$$

where $\mathcal{E}$ is the set of entity records, $\mathcal{R}$ is the set of relation records, and $\mathcal{C}$ is the set of coreference records. Each record still follows the same general format $y_i = (\ell_i, \Pi_i, R_i)$.

Record Instantiation. Let L_{Entity} denote the SciERC entity-type inventory, L_{Relation} denote the relation-type inventory, and $\mathcal{D}(s)$ denote the set of sentence identifiers in the source document. The predicted SciERC output is partitioned into entity, relation, and coreference records:

$$\widehat{Y}_{\text{SciERC}} = \mathcal{E} \,\dot\cup\, \mathcal{R} \,\dot\cup\, \mathcal{C}. \tag{E8}$$

Entity records are typed mentions grounded in the source text:

$$e = \big(t_e, \{\pi_e\}, \emptyset\big), \qquad t_e \in L_{\text{Entity}}, \quad \pi_e \in \mathcal{S}(s). \tag{E9}$$

Here, t_e is the entity type and π_e is the text span that grounds the entity mention.

Relation records are typed directed links from one grounded entity mention to another:

$$r = \big(\{\pi_{r1}\}, t_r, \{\pi_{r2}\}\big), t_r \in L_{\text{Relation}}, \pi_{r1}, \pi_{r2} \in \mathcal{S}(s). \tag{E10}$$

Here, t_r is the relation type, and the ordered pair of grounding sets $(\{\pi_{r1}\}, \{\pi_{r2}\})$ specifies the directed relation from the first grounded entity mention to the second grounded entity mention.

Coreference records link grounded entity mentions that refer to the same underlying scientific concept:

$$c = \big(\{\pi_{c1}\}, d_1, \{\pi_{c2}\}, d_2\big), \pi_{c1}, \pi_{c2} \in \mathcal{S}(s), d_1, d_2 \in \mathcal{I}_{\text{sent}}(s). \tag{E11}$$

Here, $\{\pi_{c1}\}$ and $\{\pi_{c2}\}$ are the grounding sets for the two linked mentions, and d_1 and d_2 are their corresponding sentence identifiers.

Appendix F Metric Definitions

Each task adopts an evaluation strategy tailored to its prediction format and semantic structure. This appendix defines the task-specific metrics used for PV-Miner and SciERC. PV-Miner is evaluated through hierarchical semantic-label prediction and textual grounding, using Code F1, Sub-code F1, and Span F1. SciERC is evaluated through scientific information extraction, using Entity F1, Relation F1, Span F1, and Coreference F1 to measure semantic labelling, textual grounding, and relational linking.

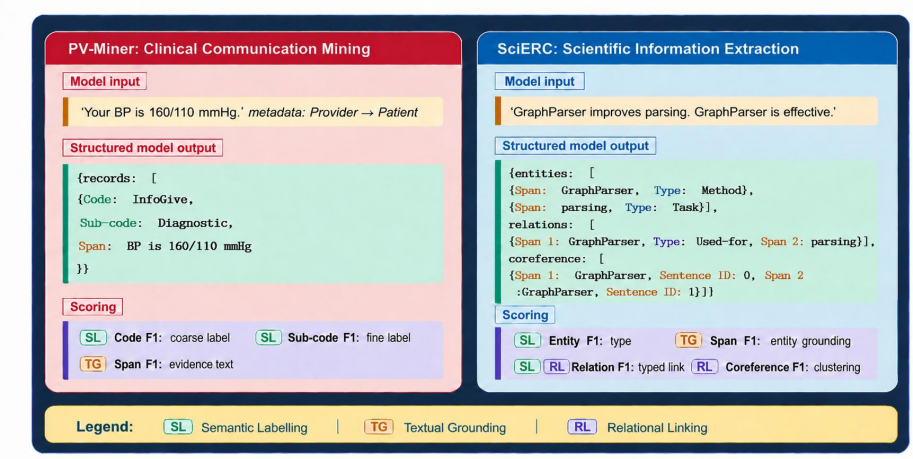


Fig. F1: Examples of the two ontology-constrained structured prediction tasks evaluated in this paper. **PV Miner** requires hierarchical clinical-communication labels with grounded evidence spans; **SciERC** requires entity typing, span grounding, relation extraction, and coreference linking..

F.1 PV-Miner

Code Classification

The Code classification task is formulated as a multi-label classification problem over a predefined set of communicative Codes. Let $\widehat{\mathcal{K}}^{(i)}$ denote the predicted Code set and $\mathcal{K}^{(i)}$ the gold Code set for instance i. We compute precision recall, and F1-score as follows:

$$\text{precision}_{\text{Code}} = \frac{\sum_i \left|\widehat{\mathcal{K}}^{(i)} \cap \mathcal{K}^{(i)}\right|}{\sum_i \left|\widehat{\mathcal{K}}^{(i)}\right|}$$

$$\text{recall}_{\text{Code}} = \frac{\sum_i \left|\widehat{\mathcal{K}}^{(i)} \cap \mathcal{K}^{(i)}\right|}{\sum_i \left|\mathcal{K}^{(i)}\right|} \tag{F1}$$

$$F1_{\text{Code}} = \frac{2 \times \text{precision}_{\text{Code}} \times \text{recall}_{\text{Code}}}{\text{precision}_{\text{Code}} + \text{recall}_{\text{Code}}}.$$

Sub-code Classification

Sub-code classification is also evaluated as a multi-label task, where each message may be annotated with one or more Sub-codes tied to a parent Code. Let $\widehat{\mathcal{V}}^{(i)}$ and $\mathcal{V}^{(i)}$ denote predicted and gold Sub-code sets, respectively. Metrics are calculated using:

$$\text{precision}_{\text{Sub-code}} = \frac{\sum_i \left|\widehat{\mathcal{V}}^{(i)} \cap \mathcal{V}^{(i)}\right|}{\sum_i \left|\widehat{\mathcal{V}}^{(i)}\right|}$$

$$\text{recall}_{\text{Sub-code}} = \frac{\sum_i \left|\widehat{\mathcal{V}}^{(i)} \cap \mathcal{V}^{(i)}\right|}{\sum_i \left|\mathcal{V}^{(i)}\right|} \tag{F2}$$

$$F1_{\text{Sub-code}} = \frac{2\,\text{precision}_{\text{Sub-code}}\,\text{recall}_{\text{Sub-code}}}{\text{precision}_{\text{Sub-code}} + \text{recall}_{\text{Sub-code}}}.$$

Span

For Span extraction, we evaluate each predicted evidence string against gold Spans using a **relaxed token-level matching strategy**, which combines:

- Full containment (i.e., gold Span is fully included in predicted Span or vice versa).
- Jaccard similarity (between predicted and gold Spans), with a threshold of 0.6.

Let $\widehat{\Pi}^{(i)}$ and $\Pi^{(i)}$ denote the sets of predicted and gold evidence Spans (strings) for instance i. A predicted Span $\hat{\pi} \in \widehat{\Pi}^{(i)}$ is counted as a **true positive (TP)** if it matches any gold Span $\pi \in \Pi^{(i)}$ such that $\text{Tok}(\pi) \subseteq \text{Tok}(\hat{\pi})$ or $\text{Tok}(\hat{\pi}) \subseteq \text{Tok}(\pi)$ or $\text{Jaccard}(\hat{\pi}, \pi) \geq 0.6$. Spans in $\widehat{\Pi}^{(i)}$ that fail to match any gold Span are counted as **false positives (FP)**, and Spans in $\Pi^{(i)}$ not matched by any prediction are counted as **false negatives (FN)**. Precision, recall, and F1-score are then computed as:

$$\text{precision}_{\text{Span}} = \frac{|\text{TP}|}{|\text{TP} + \text{FP}|}$$

$$\text{recall}_{\text{Span}} = \frac{|\text{TP}|}{|\text{TP} + \text{FN}|} \tag{F3}$$

$$F1_{\text{Span}} = \frac{2 \times \text{precision}_{\text{Span}} \times \text{recall}_{\text{Span}}}{\text{precision}_{\text{Span}} + \text{recall}_{\text{Span}}}.$$

F.2 SciERC

Entity

The Entity classification task is formulated as a multi-label classification problem over a predefined set of scientific entity types. Let $\widehat{\mathcal{T}}^{(i)}_{\text{Entity}}$ denote the predicted entity-type set for instance i, and let $\mathcal{T}^{(i)}_{\text{Entity}}$ denote the corresponding gold-standard entity-type set, where these sets are obtained by projecting each entity record onto its label component t_e. We compute precision, recall, and F1-score as follows:

$$\text{precision}_{\text{Entity}} = \frac{\sum_i \left|\widehat{\mathcal{T}}^{(i)}_{\text{Entity}} \cap \mathcal{T}^{(i)}_{\text{Entity}}\right|}{\sum_i \left|\widehat{\mathcal{T}}^{(i)}_{\text{Entity}}\right|}$$

$$\text{recall}_{\text{Entity}} = \frac{\sum_i \left|\widehat{\mathcal{T}}^{(i)}_{\text{Entity}} \cap \mathcal{T}^{(i)}_{\text{Entity}}\right|}{\sum_i \left|\mathcal{T}^{(i)}_{\text{Entity}}\right|} \tag{F4}$$

$$F1_{\text{Entity}} = \frac{2\,\text{precision}_{\text{Entity}}\,\text{recall}_{\text{Entity}}}{\text{precision}_{\text{Entity}} + \text{recall}_{\text{Entity}}}.$$

Span

For Entity Span extraction, we evaluate each predicted evidence string against gold Entity Spans using a **relaxed token-level matching strategy**, which combines:

- Full containment, i.e., the gold Span is fully included in the predicted Span or vice versa.
- Jaccard similarity between predicted and gold Spans, with a threshold of 0.6.

Let $\widehat{\Pi}^{(i)}_e$ and $\Pi^{(i)}_e$ denote the sets of predicted and gold evidence Spans for instance i, where each grounding set satisfies $\Pi^{(i)}_e \subseteq \mathcal{S}(s_i)$ and anchors the structured record to the source sequence s_i.

For a predicted Span $\hat{\pi_e} \in \widehat{\Pi_e}^{(i)}$ and a gold Span $\pi_e \in \Pi^{(i)}_e$, let $\text{Tok}(\cdot)$ denote the set of tokens in the Span. The relaxed Span-matching function is defined as:

$$\begin{aligned}\text{Match}_{\Pi_e}(\hat{\pi}_e, \pi_e) = &\mathbb{I}\,[\text{Tok}(\pi_e) \subseteq \text{Tok}(\hat{\pi}_e) \vee \\ &[0.15em]\text{Tok}(\hat{\pi}_e) \subseteq \text{Tok}(\pi_e) \;\vee\; \text{Jaccard}(\hat{\pi}_e, \pi_e) \geq 0.6]\end{aligned} \tag{F5}$$

A predicted Span $\hat{\pi} \in \widehat{\Pi}^{(i)}$ is counted as a **true positive (TP)** if it matches any gold Span $\pi \in \Pi^{(i)}$ under Match_Π. Spans in $\widehat{\Pi}^{(i)}$ that fail to match any gold Span are counted as **false positives (FP)**, and Spans in $\Pi^{(i)}$ not matched by any prediction are counted as **false negatives (FN)**. Precision, recall, and F1-score are then computed as:

$$\text{precision}_{\text{Span}} = \frac{|\text{TP}_{\text{Span}}|}{|\text{TP}_{\text{Span}}| + |\text{FP}_{\text{Span}}|}$$

$$\text{recall}_{\text{Span}} = \frac{|\text{TP}_{\text{Span}}|}{|\text{TP}_{\text{Span}}| + |\text{FN}_{\text{Span}}|} \tag{F6}$$

$$F1_{\text{Span}} = \frac{2\,\text{precision}_{\text{Span}}\,\text{recall}_{\text{Span}}}{\text{precision}_{\text{Span}} + \text{recall}_{\text{Span}}}.$$

Relation

Relation extraction is evaluated by jointly checking the relation type and the ordered grounding Spans for the two relation arguments. Let a predicted relation record be represented as $\hat{r} = (\widehat{\Pi}_{r1}^{(i)}, \hat{t}, \widehat{\Pi}_{r2}^{(i)})$, where $\widehat{\Pi}_{r1}^{(i)}$ and $\widehat{\Pi}_{r2}^{(i)}$ are the predicted grounding sets for the first and second relation arguments, and $\hat{t}$ is the predicted relation type. Similarly, let a gold relation record be represented as $r = (\Pi_{r1}^{(i)}, t, \Pi_{r2}^{(i)})$.

A predicted relation is considered correct only if the relation type matches and both argument grounding sets match their corresponding gold grounding sets in the correct order:

$$\begin{aligned}\text{Match}_{\text{Relation}}(\hat{r}, r) = \mathbb{I}\Big[\hat{t} = t \wedge\ \exists\, \hat{\pi}_{r1} \in \widehat{\Pi}_{r1}^{(i)}, \pi_{r1} \in \Pi_{r1}^{(i)} \\ \text{s.t. } \text{Match}_{\Pi}(\hat{\pi}_{r1}, \pi_{r1}) = 1 \wedge\ \exists\, \hat{\pi}_{r2} \in \widehat{\Pi}_{r2}^{(i)}, \pi_{r2} \in \Pi_{r2}^{(i)} \\ \text{s.t. } \text{Match}_{\Pi}(\hat{\pi}_{r2}, \pi_{r2}) = 1\Big].\end{aligned} \tag{F7}$$

The ordering of relation arguments is strict: the predicted first argument grounding set $\widehat{\Pi}_{r1}^{(i)}$ must match the gold first argument grounding set $\Pi_{r1}^{(i)}$, and the predicted second argument grounding set $\widehat{\Pi}_{r2}^{(i)}$ must match the gold second argument grounding set $\Pi_{r2}^{(i)}$.

After one-to-one matching between predicted and gold relation records, precision, recall, and F1-score are computed as:

$$\begin{aligned}\text{precision}_{\text{Relation}} &= \frac{|\text{TP}_{\text{Relation}}|}{|\text{TP}_{\text{Relation}}| + |\text{FP}_{\text{Relation}}|} \\ \text{recall}_{\text{Relation}} &= \frac{|\text{TP}_{\text{Relation}}|}{|\text{TP}_{\text{Relation}}| + |\text{FN}_{\text{Relation}}|} \\ F1_{\text{Relation}} &= \frac{2\,\text{precision}_{\text{Relation}}\,\text{recall}_{\text{Relation}}}{\text{precision}_{\text{Relation}} + \text{recall}_{\text{Relation}}}.\end{aligned} \tag{F8}$$

Coreference

Coreference is evaluated at the coreference-link level. Let a predicted coreference link be represented as $\hat{c} = (\widehat{\Pi}_{c1}^{(i)}, \hat{d}_1, \widehat{\Pi}_{c2}^{(i)}, \hat{d}_2)$, where $\widehat{\Pi}_{c1}^{(i)}$ and $\widehat{\Pi}_{c2}^{(i)}$ are the predicted grounding sets for the two linked mentions, and $\hat{d}_1$ and $\hat{d}_2$ are their predicted sentence identifiers. Similarly, let a gold coreference link be represented as $c = (\Pi_{c1}^{(i)}, d_1, \Pi_{c2}^{(i)}, d_2)$.

A predicted coreference link is considered correct only if both mention grounding sets match under relaxed Span matching and both sentence identifiers match exactly:

$$\mathrm{Match}_{\mathrm{Coref}}(\hat{c}, c) = \mathbb{I}\Big[\exists \hat{\pi}_{c1} \in \widehat{\Pi}_{c1}^{(i)},\ \pi_{c1} \in \Pi_{c1}^{(i)} : \mathrm{Match}_{\Pi}(\hat{\pi}_{c1}, \pi_{c1}) = 1 \ \wedge\ \hat{d}_1 = d_1 \\ \wedge\ \exists \hat{\pi}_{c2} \in \widehat{\Pi}_{c2}^{(i)},\ \pi_{c2} \in \Pi_{c2}^{(i)} : \mathrm{Match}_{\Pi}(\hat{\pi}_{c2}, \pi_{c2}) = 1 \ \wedge\ \hat{d}_2 = d_2\Big]. \tag{F9}$$

Thus, coreference evaluation requires relaxed textual agreement for both linked mention grounding sets and strict agreement for their sentence identifiers.

After one-to-one matching between predicted and gold coreference links, precision, recall, and F1-score are computed as:

$$\begin{aligned} \mathrm{precision}_{\mathrm{Coreference}} &= \frac{|\mathrm{TP}_{\mathrm{Coreference}}|}{|\mathrm{TP}_{\mathrm{Coreference}}| + |\mathrm{FP}_{\mathrm{Coreference}}|} \\ \mathrm{recall}_{\mathrm{Coreference}} &= \frac{|\mathrm{TP}_{\mathrm{Coreference}}|}{|\mathrm{TP}_{\mathrm{Coreference}}| + |\mathrm{FN}_{\mathrm{Coreference}}|} \\ F1_{\mathrm{Coreference}} &= \frac{2\,\mathrm{precision}_{\mathrm{Coreference}}\,\mathrm{recall}_{\mathrm{Coreference}}}{\mathrm{precision}_{\mathrm{Coreference}} + \mathrm{recall}_{\mathrm{Coreference}}}. \end{aligned} \tag{F10}$$

Appendix G Related Work

Preference-based alignment and offline preference optimization.

Preference-based alignment has become a standard post-supervised fine-tuning strategy for adapting large language models to human or task-specific preferences. Early RLHF pipelines learn a reward model and optimize the policy with online reinforcement learning, but these pipelines require sampling during training and can be computationally expensive and difficult to stabilize [8, 10]. Direct Preference Optimization (DPO) simplifies this pipeline by replacing explicit reward modeling and online RL with an offline, reference-regularized classification objective over preferred and rejected completions [9]. This formulation is effective for many open-ended generation settings, but it remains fundamentally sequence-level: the preference signal is assigned to entire completions rather than to the few tokens that determine structured correctness.

Sequence-level DPO variants.

A large body of recent work improves DPO by modifying the sequence-level preference objective. IPO and the broader ΨPO framework analyze preference optimization from a theoretical perspective and replace the logistic classification target with a more stable margin-regression objective [14]. SimPO removes the explicit reference model and uses average sequence log-likelihood as an implicit reward, improving simplicity and length behavior [15]. Cal-DPO argues that contrastive preference objectives can ignore the absolute scale of implicit rewards and introduces calibration to make learned rewards more comparable to target rewards [16]. DPO-Positive / DPOP addresses

the observation that DPO can increase the relative preference margin while degrading the likelihood of the chosen response, adding a correction that protects preferred completions [17]. These methods address important sequence-level pathologies such as reference dependence, margin instability, calibration drift, and chosen-likelihood degradation. However, they still operate primarily through completion-level scalar margins. In ontology-driven structured prediction, this is insufficient because the preferred and rejected outputs are often near-identical serialized objects whose correctness differs in only a small number of schema-defining tokens.

Low-edit-distance structured preferences.

Structured extraction tasks differ from open-ended alignment tasks because small local errors can invalidate an otherwise well-formed output. In PV-Miner, a single Code or Sub-code token can change the clinical-communication interpretation of a grounded span. In SciERC, a single entity type, relation label, sentence identifier, or coreference link can determine whether a structured record is correct. These tasks therefore induce low-edit-distance preference pairs: the chosen and rejected completions share most JSON scaffolding, field names, punctuation, and repeated formatting tokens, while differing only at sparse semantic decision points. Under standard DPO, the post-divergence gradient can be spread across both critical and non-critical tokens. This creates gradient dilution, where optimization mass is spent on serialization tokens that do not determine task F1, and preferred-token erosion, where the aggregate preference margin can improve even as the likelihood of rare but correct schema tokens decreases. TAB-PO is designed for this regime rather than for generic response-level preference separation.

Token-level preference optimization.

Token-level preference optimization methods move beyond purely completion-level objectives. TDPO decomposes preference optimization at the token level and introduces token-wise forward KL constraints to improve the balance between alignment and generation diversity [18]. TI-DPO further argues that tokens differ in importance and estimates token-importance weights using gradient attribution, together with an additional triplet-style loss that pulls the policy toward preferred responses and away from rejected responses [19]. These approaches are closer to TAB-PO than sequence-level DPO variants because they recognize that token positions contribute unequally to preference learning. However, their token-level mechanisms are not specialized for ontology-constrained structured outputs. TDPO is primarily a token-level KL reformulation for general alignment, while TI-DPO infers token importance from model-derived attribution signals. TAB-PO instead uses the structure of the task itself: it constructs minimally perturbed, schema-valid rejected outputs from empirical SFT confusions, and it applies a confidence-gated barrier only to preferred tokens whose current likelihood falls below a threshold. This distinction is reflected in the training diagnostics: TAB-PO achieves stronger preference-probability convergence while maintaining a more stable reference-adjusted margin, rather than increasing the sequence-level margin by sacrificing under-confident schema tokens. Thus, TAB-PO

does not merely ask which tokens are important in general; it asks which schema-defining preferred tokens are currently at risk of probability erosion and protects those tokens during preference learning.

Verifier-based RL and GRPO-style alternatives.

Recent reasoning-oriented alignment methods use reinforcement learning with verifiable rewards, including GRPO-style optimization for mathematical reasoning [31]. More generally, RLVR is effective when candidate outputs can be scored by reliable programmatic or binary verifiers, such as final-answer checks in mathematics, executable tests in code, or other domains with objective outcome-level rewards [32]. Ontology-driven clinical and scientific annotation has a different supervision structure. The gold output is available, but the clinically or semantically meaningful distinction is often local, hierarchical, and ambiguity-sensitive rather than reducible to a single scalar reward. In this setting, expert-adjudicated and confusion-aware preference pairs provide a more targeted signal than online scalar rewards: they encode exactly which structured alternative should be preferred and preserve the low-edit-distance contrast that exposes label, span, relation, and coreference errors. TAB-PO also has a practical computational advantage: after supervised fine-tuning, its preference data can be prepared at near-zero marginal cost from existing gold records, SFT validation errors, and empirical confusion tables, without online rollout sampling, reward-model training, external verifier execution, or group-based policy optimization. TAB-PO is therefore complementary to verifier-based RL, but it is tailored to the offline post-SFT setting where gold structured records and empirical model confusions are already available.

Positioning of TAB-PO.

TAB-PO differs from prior preference-optimization methods along two axes. First, its preference data are confusion-aware: rejected outputs are minimally perturbed, schema-valid alternatives sampled from the SFT model's empirical ontology-level confusions and expert-observed ambiguity patterns. This makes optimization focus on realistic structured errors rather than arbitrary negative completions. Second, its objective is token-critical: the confidence-gated barrier restores likelihood only for under-confident preferred tokens, preserving the *critical schema-defining tokens* that determine structured correctness, including labels, spans, relation arguments, sentence identifiers, and coreference links. This protection is localized: confident tokens can still support preference separation, while under-confident semantic tokens are prevented from eroding below a useful likelihood floor. This combination directly addresses the two failure modes that arise in low-edit-distance structured preferences: gradient dilution over non-critical serialization tokens and erosion of rare preferred schema tokens. As a result, TAB-PO is not simply another DPO variant for general alignment; it is a post-SFT preference objective specialized for ontology-constrained structured generation.

Appendix H Targeted TAB-PO Negative Construction

This appendix illustrates how TAB-PO constructs targeted negative outputs for ontology-driven structured prediction. Given an input x and gold structured output Y^+, TAB-PO constructs a rejected output Y^- by introducing a small, schema-valid perturbation that reflects a realistic structured prediction error. The goal is not to create arbitrary corrupted outputs, but to produce low-separation hard negatives: outputs that remain parseable and ontology-valid while differing from the gold output in the semantic label, textual grounding, or relational-linking decision that determines correctness. Figures H1, H2, and H3 below illustrate representative perturbations for PV-Miner and SciERC.

Targeted TAB-PO negative construction for PV Miner

Legend: Semantic labelling confusion Missing record perturbation Extra record perturbation

Metadata: TO_PAT_YN=Y *(Provider→Patient)*

Input message: *No worries! Thank you for letting me know. Best wishes, Dr. Person1*

Ground truth:

```
[
  {Code: PartnershipProvider, Sub-code: connection, Span: "No worries!"},
  {Code: PartnershipProvider, Sub-code: Appreciation/Gratitude,
                              Span: "Thank you for letting me know"},
  {Code: PartnershipProvider, Sub-code: signoff, Span: "Best wishes"}
]
```

1) Semantic labelling confusion

A semantic label is perturbed while the grounded span is retained. Code–Sub-code mapping is preserved under Semantic labelling confusion.

```
[
  {Code: PartnershipPatient, Sub-code: connection, Span: "No worries!"},
  {Code: PartnershipProvider, Sub-code: signoff,
                              Span: "Thank you for letting me know"},
  {Code: PartnershipProvider, Sub-code: signoff, Span: "Best wishes"}
]
```

2) Missing record perturbation

A valid ground-truth record is removed from the structured output.

```
[
  {Code: PartnershipProvider, Sub-code: connection, Span: "No worries!"},
  {Code: PartnershipProvider, Sub-code: Appreciation/Gratitude,
                              Span: "Thank you for letting me know"}
]
```

Removed: `{Code: PartnershipProvider, Sub-code: signoff, Span: "Best wishes"}`

3) Extra record perturbation

An additional validly formatted but unsupported record is inserted. The Span for the extra record is chosen from the input message.

```
[
  {Code: PartnershipProvider, Sub-code: connection, Span: "No worries!"},
  {Code: PartnershipProvider, Sub-code: Appreciation/Gratitude,
                              Span: "Thank you for letting me know"},
  {Code: PartnershipProvider, Sub-code: signoff, Span: "Best wishes"},
  {Code: SDOH, Sub-code: SocialAndCommunityContext, Span: "No worries"}
]
```

Fig. H1: PV-Miner example of targeted TAB-PO negative construction.

Targeted TAB-PO negative construction for SciERC

Legend: Semantic labelling confusion Relational linking confusion Missing record perturbation Extra record perturbation

Input sample: *GraphParser improves parsing. GraphParser is effective.*

Ground-truth:

```
{
  entities: [
    {Span: GraphParser, Type: Method},
    {Span: parsing, Type: Task}
  ],
  relations: [
    {Span 1: GraphParser, Type: Used-for, Span 2: parsing}
  ],
  coreference: [
    {Span 1: GraphParser, Sentence ID: 0,
     Span 2: GraphParser, Sentence ID: 1}
  ]
}
```

1) Semantic labelling confusion

A semantic label is perturbed while the grounded span and structured record are retained. For SciERC, this can perturb an entity type, or relation type.

```
{
  entities: [
    {Span: GraphParser, Type: Method},
    {Span: parsing, Type: Method}
  ],
  relations: [
    {Span 1: GraphParser, Type: Evaluate-for, Span 2: parsing}
  ],
  coreference: [
    {Span 1: GraphParser, Sentence ID: 0,
     Span 2: GraphParser, Sentence ID: 1}
  ]
}
```

2) Relational linking confusion

A relational link is perturbed while the spans remain grounded in the input. This may involve reversing Span 1 and Span 2, confusing one linked span with another entity span, or modifying a coreference link by substituting another entity span with a valid sentence ID.

```
{
  entities: [
    {Span: GraphParser, Type: Method},
    {Span: parsing, Type: Task}
  ],
  relations: [
    {Span 1: parsing, Type: Used-for, Span 2: GraphParser}
  ],
  coreference: [
    {Span 1: parsing, Sentence ID: 0,
     Span 2: GraphParser, Sentence ID: 1}
  ]
}
```

Fig. H2: SciERC semantic-label and relational-link confusion examples for TAB-PO negative construction.

3) Missing record perturbation

A valid ground-truth record is removed from the structured output. For SciERC, this may involve removing an entity, relation, or coreference link; if an entity is removed, its associated relations and coreference links are also removed to maintain structural validity.

```
{
  entities: [
    {Span: GraphParser, Type: Method}
  ],
  relations: [
    {Span 1: GraphParser, Type: Used-for, Span 2: parsing}
  ],
  coreference: [
    {Span 1: GraphParser, Sentence ID: 0,
     Span 2: GraphParser, Sentence ID: 1}

]
}
```

Removed: {Span: parsing, Type: Task}

4) Extra record perturbation

An additional validly formatted but unsupported record is inserted. For entity-level extra records, the span is chosen from the input text.

```
{
  entities: [
    {Span: GraphParser, Type: Method},
    {Span: parsing, Type: Task},
    {Span: effective, Type: Metric}
  ],
  relations: [
    {Span 1: GraphParser, Type: Used-for, Span 2: parsing}
  ],
  coreference: [
    {Span 1: GraphParser, Sentence ID: 0,
     Span 2: GraphParser, Sentence ID: 1}
  ]
}
```

Fig. H3: SciERC missing-record and extra-record perturbation examples for TAB-PO negative construction.

PV-Miner focuses on hierarchical clinical annotation, where negative construction targets Code/Sub-code confusions, missing grounded records, and unsupported extra records. SciERC extends the same idea to scientific relational extraction, where perturbations can affect entity labels, relation labels, relation arguments, coreference links, missing records, and extra records. Across all perturbation types, the rejected output remains close to the preferred output while changing the specific ontology-bound decision that determines correctness. TAB-PO is designed to overcome the disadvantage of low-separation structured preferences: it converts preference optimization into a targeted signal for schema-critical tokens rather than allowing the update to be diluted across shared serialization structure.